Cluster-Aware Uncertainty for Polytomous Mokken Scalability: A Calibrated Hierarchical Bayesian Bootstrap

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ABSTRACT. Mokken scale analysis condenses whether a fixed set of items forms a scale into a single scalability coefficient, and much of the data it serves is collected in groups: pupils within classrooms, patients within clinics. Yet the coefficient's uncertainty is still routinely computed as if respondents were independent. Such intervals are shorter than the evidence allows and miss the true coefficient far more often than their label promises, while looking all the more precise; and once groups differ in size, "the population coefficient" is itself ambiguous, because the value for the average respondent and the value for the average group need not coincide. We develop confidence intervals for the polytomous scale coefficient that repair both problems. The two population versions are defined separately, and every interval is labeled by the one it estimates. Uncertainty comes from a hierarchical Bayesian bootstrap that reweights groups and respondents within groups, with the coefficient computed exactly under every reweighting; because the resulting interval overshoots its advertised coverage, it is tightened by a deterministic factor whose single constant was fixed on training conditions, validated on held-out conditions, and locked before the final evaluation existed. In a demanding confirmatory simulation the calibrated interval held its advertised coverage for both population targets at a fraction of the uncorrected width; applied without retuning, it delivers stable cluster-aware intervals for two scales of pupils' school well-being. The procedure is implemented in the R package bayesmokken, and we suggest that a scalability claim from grouped data should routinely carry such an interval, labeled by its target.

Keywords: Mokken scale analysis; multilevel data; Bayesian bootstrap; scalability coefficient; polytomous items

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1. **Introduction**

Much of the questionnaire data collected in psychology is nested: pupils respond within classrooms, patients within clinics, and employees within teams. Most of the scales constructed from such data are also polytomous, consisting of ordered category responses rather than binary ones. Nevertheless, when researchers ask whether a set of items orders respondents consistently enough to be scored together, the standard answer, Mokken scale analysis with its scalability coefficient H (Mokken, 1971; Molenaar, 1991; Sijtsma & Molenaar, 2002), is still predominantly computed and interpreted as if respondents were independent. The consequences are not negligible. In the school data analyzed below, classroom membership accounts for about one sixth of the latent variation, and our simulations show what this does to the standard machinery: an interval that reweights respondents as if they were independent covers a nominal 95% target as little as 88% of the time, while its cluster-blind length makes it appear misleadingly precise. Nothing in the interval itself reveals the failure, which is why it has to be prevented rather than detected.

The two-level Mokken literature has developed substantial machinery for this setting. Koopman, Zijlstra, and van der Ark (2020) derived standard errors for two-level scalability coefficients, Koopman, Zijlstra, de Rooij, and van der Ark (2020) documented their small-sample bias, and the two-step, test-guided workflow of Koopman et al. (2022), with model-fit machinery in Koopman et al. (2023) and theoretical foundations in Koopman et al. (2025), has made clustered Mokken analysis practicable. This article builds directly on that work and addresses three questions that it leaves open. The first concerns the estimand: with unequal cluster sizes, "the population coefficient" is ambiguous, since a respondent-weighted population mixture and an equal-cluster mixture are different quantities whenever cluster size is informative (Nevalainen et al., 2014; Williamson et al., 2003), and an uncertainty statement should specify which of the two it concerns. The second concerns calibration: analytic standard errors yield symmetric Wald intervals whose finite-sample coverage for this bounded, kinked ratio statistic had not been examined under prespecified rules. The third concerns the inferential engine: resampling-free Bayesian bootstrap (BB) weighting, which handles the coefficient's nondifferentiable structure without derivatives when respondents are independent, had no established clustered version, and we show that the two obvious adaptations fail in opposite directions.

Our proposal is a two-stage hierarchical Bayesian bootstrap (HBB) with an explicit width calibration at the cluster rate. Clusters receive independent Dirichlet weights, respondents receive independent Dirichlet weights within clusters, and the two layers combine into draw weights whose algebra exactly targets either the respondent-weighted or the equal-cluster population coefficient (for the broader family of weighted and hierarchical BB constructions, see Lo, 1987; Makela et al., 2018; Oganisian et al., 2024; Rubin, 1981; Zhou et al., 2016). The raw two-stage interval turns out to need correcting in the direction opposite to its independent-respondent counterpart: whereas the binary respondent-level BB interval undercovers slightly and must be widened (Lee, 2026b), the clustered two-stage interval overcovers substantially, consistent with two independent weighting stages injecting more variability than the sampled clusters warrant. The correction is again the smallest one we could justify: the interval is shrunk about its midpoint by a deterministic factor that falls short of one by a fixed constant divided by the square root of the number of clusters. The constant was fixed on training data, selected among candidates that survived frozen eligibility screens, validated on disjoint held-out conditions, locked, and only then evaluated, so it was never tuned on the data that judge it. Neither this constant nor the binary one transfers to the other setting; the sign contrast between them is an empirical finding about the two designs, and Section 7 returns to what it suggests about weight-law calibration in general.

The evidence was assembled in the manner of a confirmatory study. In a fresh 24-cell design never touched during development, crossing 22 to 80 clusters with latent intraclass correlations from .06 to .38, four cluster-size mechanisms including informative sizes in both directions, and deliberate transport-path stress lanes, the locked interval covered .974 for both population targets on the 16-cell regular decision lane, inside the prespecified bands, with all seven blocking gates passing and at three-quarters of the raw interval's width. The naive independence BB covered .915 on the same lane, and the strongest available baseline, the two-level delta interval of Koopman, Zijlstra, and van der Ark (2020), covered .981 at 7.1% greater length than ours. We then applied the frozen procedure, without retuning, to the prespecified SWMDK school data of 639 pupils in 30 classrooms (Koopman et al., 2022), where it delivers stable, cluster-aware intervals for both well-being scales, labeled by target, at just under three-quarters of the raw width. The complete procedure is implemented in the bayesmokken R package.

The article makes five contributions: an estimand-first formalization of clustered polytomous scalability with two explicitly separated population targets; an exact finite-transport computation of the weighted coefficient in the classical lineage of

Molenaar (1991) and Rüschendorf (1982), with prior weighted software by Andreadis (2017) credited explicitly; a two-stage hierarchical weight law with target-exact algebra and a calibrated cluster-rate width correction whose selection never saw the data that judged it; a fresh confirmatory evaluation together with a prespecified real-data application with full diagnostic disclosure; and software with a complete replication chain. The intervals are BB-based confidence intervals evaluated by repeated-sampling coverage, and the evidence is specific to the frozen designs; Section 7 states the boundaries in full.

The plan of the article is as follows. Section 2 reviews the weighted polytomous coefficient, the two-level Mokken literature, and cluster resampling. Section 3 defines the two population targets, describes the exact transport computation, states the hierarchical weight law with its exact identities, and develops the width calibration. Section 4 gives the algorithm, the selection chronology of the constant, and the software, with a transparency subsection recording data, code, and locking. Section 5 and Section 6 report the confirmatory simulation and the SWMDK application, and Section 7 concludes. Proofs, exact algorithm and comparator specifications, data-generating details, the governance chronology, the complete confirmation tables, and the application details are collected in the online supplement, Appendices A–H of this document.

2. Background

2.1. The Weighted Polytomous Scalability Coefficient. For binary items, Loevinger's H compares observed item covariances with the maximum attainable under the margins. Molenaar (1991) extended the coefficient to multicategory items: ordered categories are scored, Guttman errors are weighted by the number of reversed item steps, and — in the covariance form we compute — the pair numerator is the covariance of two ordinal scores while the pair denominator is the maximum covariance compatible with the two score margins, with item and scale coefficients aggregating numerators and denominators before dividing (ratio of sums, not mean pair coefficient; Molenaar's Theorems 2 and 5 establish the equivalence of the weighted-error and covariance-ratio objects). The margin-constrained maximum itself belongs to an older lineage: for fixed marginals, the common-quantile (comonotone) coupling maximizes the expectation of convex functions of sums (Rüschendorf, 1982); specializing to two variables and the squared sum yields exactly the maximum-covariance coupling the denominator requires (see also Puccetti & Wang, 2015). The coefficient family sits at the center of applied nonparametric IRT practice (Sijtsma & Molenaar, 2002; Sijtsma & van der Ark, 2017; Sijtsma et al., 2011), with polytomous foundations for the underlying monotone models in Hemker et al. (1996, 1997) and graded-response parameterizations tracing to Samejima (1969).

The software record should be stated plainly: weighted Mokken scale analysis in R predates the present work. Andreadis (2017) released a weighted implementation of the coefficient for survey-weighted data; our implementation differs in its validated weight interface, declared score supports, and typed missing-data contract, but the idea of computing Mokken coefficients under respondent weights is his precedent, and nothing in this article should be read as a first-implementation claim. What this article adds on the computational side is narrower and stated exactly in the theory section: an arbitrary-mass finite transport evaluation of the denominator that remains exact under the fractional, data-dependent weights our inference engine generates.

2.2. **Two-Level Mokken Analysis.** Clustered administration is the norm in the settings where Mokken analysis is most used (schools, clinics, and teams), and a dedicated two-level literature now exists. Koopman et al. (2025) lay out nonparametric two-level IRT models and their properties; Koopman, Zijlstra, and van der Ark (2020) derive standard errors for two-level scalability coefficients (implemented in mokken's `MLcoefH`); Koopman, Zijlstra, de Rooij, and van der Ark (2020) characterize finite-sample bias in the coefficients and their standard errors; Koopman et al. (2022) give the applied workflow and distribute the SWMDK example data we adopt; and Koopman et al. (2023) add model-fit evaluation. Ratings and other nested designs raise the same structure elsewhere in measurement (Koopman & Braeken, 2025; Wind & Patil, 2018). Our two-level delta comparator is precisely this literature's interval: the coefficient plus-minus a normal quantile times the analytic two-level standard error. It is a strong baseline (in our confirmatory design it covers between .967 and .981), and our aim is not to displace the analytic route but to supply what it does not provide: an explicit choice of population target, finite-G calibration evidence under prespecified gates, and an engine that handles the coefficient's kinks without derivatives.

One more ingredient from the sampling literature is essential. With unequal cluster sizes, respondent-level and cluster-level summaries answer different questions, and when size is *informative* (that is, correlated with the outcome-generating process) the two population mixtures genuinely diverge (Nevalainen et al., 2014; Williamson et al., 2003). Scale coefficients are no exception. We therefore carry two estimands through the entire article rather than allowing a weighting convention to select one implicitly.

2.3. **Cluster Resampling and Hierarchical Bayesian Weighting.** The frequentist cluster bootstrap resamples clusters (and possibly respondents within them), with consistency results for clustered estimating equations (Cheng et al., 2013) and practical guidance across fields (Deen & de Rooij, 2020; Field & Welsh, 2007; Huang, 2018); within-cluster resampling offers an alternative resolution of informative size (Hoffman et al., 2001). On the Bayesian side, Rubin's (1981) Dirichlet weighting has first-order equivalence theory (Lo, 1987) and finite-population versions (Lo, 1988); two-stage and finite-population extensions weight design stages hierarchically (Dong et al., 2014; Zhou et al., 2016), Bayesian inference under cluster sampling with unequal probabilities is developed in Makela et al. (2018), and hierarchical BB variants appear in modern causal applications (Oganisian et al., 2024). For binary scales with independent respondents, Lee (2026b) developed a width-calibrated BB confidence interval for the scale coefficient under the same governed train–validate–confirm design that this article uses. None of this work, however, supplies the combination that a clustered scalability coefficient requires: draw weights whose algebra exactly matches a declared cluster estimand of a kinked ratio functional, together with finite-G coverage calibration validated under outcome-blind rules. The following three sections assemble and evaluate that combination; throughout, every interval is a BB-based confidence interval evaluated by repeated-sampling coverage, never a posterior credible interval.

3. Estimands, Computation, and the Calibrated Interval

3.1. **Two Population Targets for Clustered Scales.** Let a fixed, prespecified set of J polytomous items with declared, strictly increasing numeric score supports be administered to respondents nested in clusters. Clusters $g = 1, \ldots, G$ are the independent sampling units, drawn from a superpopulation together with their sizes m_g and within-cluster response distributions F_g; respondents are exchangeable within clusters given (m_g, F_g); $N = \sum_g m_g$. Because (m_g, F_g) may be dependent (larger clusters may respond systematically differently), two population response distributions must be distinguished. Writing (M, F) for one superpopulation draw of a cluster's size and response distribution,

$$F_{\mathrm{R}} = \frac{\mathbb{E}(MF)}{\mathbb{E}(M)} \quad \text{(respondent-weighted)}, \qquad F_{\mathrm{C}} = \mathbb{E}(F) \quad \text{(equal-cluster)}: \quad (1)$$

the size-biased mixture is the distribution of a randomly sampled *respondent*; the unweighted mixture is that of a random respondent from a randomly sampled *cluster*. Their finite-sample plug-ins reweight the G sampled clusters accordingly ($\sum_g m_g F_g / \sum_g m_g$ and $G^{-1} \sum_g F_g$; that is, point weight $1/N$ or $1/(Gm_g)$ per respondent), but the population functionals in Equation (1), not the plug-ins, are the estimands. Under informative cluster size the two differ, and so do the scalability coefficients they induce (Nevalainen et al., 2014; Williamson et al., 2003). Our primary estimand is $H(F_{\mathrm{R}})$; $H(F_{\mathrm{C}})$ is carried throughout as a distinct sensitivity estimand, never as a substitute. Every interval in this article is labeled by its target.

For any response distribution F with score margins F_j, the weighted polytomous coefficient is

$$H(F) \;=\; \frac{\sum_{j<k} \mathrm{Cov}_F(Y_j, Y_k)}{\sum_{j<k} \mathrm{Cov}^{\max}(F_j, F_k)}, \tag{2}$$

where $\mathrm{Cov}^{\max}(F_j, F_k)$ is the largest covariance attainable by *any* joint distribution with margins F_j and F_k (Molenaar, 1991). Molenaar's presentation scores categories with consecutive integers (any equidistant scoring being equivalent); admitting arbitrary declared, strictly increasing numeric scores is a scope extension of the same covariance-ratio coefficient, with which it coincides on consecutive scores. Item coefficients replace the double sum with the sum over pairs containing item j; all aggregation is ratio-of-sums. The item set is fixed before inference; item selection, deletion, and re-keying are prohibited by protocol.

3.2. **Exact Computation by Finite Comonotone Transport.** The denominator is a classical object: for fixed margins, the common-quantile (comonotone) coupling attains the maximum expectation of a convex function of the sum (Rüschendorf, 1982, p. 624, eq. 2); with two variables and $\psi(s) = s^2$, fixed marginal second moments make maximizing $\mathbb{E}(Y_j + Y_k)^2$ equivalent to maximizing the covariance, which is a derived specialization of an established result rather than a new theorem (Appendix B). On finite supports the coupling is computed exactly by an increasing *two-pointer transport*: order both supports, advance through both margins from the lowest scores upward, and at each step ship the smaller remaining mass, accumulating the cross-moment (at most $K+L-1$ positive allocations for margin support sizes K and L, since every step exhausts at least one support point). The construction accepts arbitrary nonnegative masses (the fractional, data-dependent weights that our inference engine produces), not only multiples of $1/n$, so that the estimator is exactly evaluable on every weighted draw. Ties in the cumulative margins can make the transport *table* non-unique without changing the attained maximum; the diagnostic consequences of such *cumulative knots* are handled below. The implementation is verified against an independent linear-programming oracle (25 fixtures; maximum absolute discrepancy 1.4×10^{-12}) and 50 independent quantile-coupling fixtures, and on binary supports it reduces exactly to the classical binary computation.

Two regularity notions organize the developments that follow. A weighted-margin configuration is *regular* when the cumulative margins of every item pair are separated at all interior support points (the transport path is locally constant under perturbation); it lies at an *exact knot* when interior cumulative margins coincide, so an arbitrarily small perturbation can switch the path (with *near-knot* configurations in between). Exact knots are the polytomous analog of the binary marginal-tie problem: the functional remains well defined and exactly computable, but smooth local analysis is not guaranteed there. By frozen contract, knot configurations are *diagnostic lanes*: we report them, we do not claim ordinary first-order inference at them (Appendix A states both notions formally).

3.3. **The Two-Stage Hierarchical Weight Law.** Inference perturbs the empirical distribution with weights that respect the sampling structure. Draw, independently,

$$\begin{aligned} (A_1, \dots, A_G) &\sim \mathrm{Dirichlet}_G(1, \dots, 1), \\ (B_{g1}, \dots, B_{gm_g}) &\sim \mathrm{Dirichlet}_{m_g}(1, \dots, 1) \text{ within each cluster,} \end{aligned} \tag{3}$$

and combine them into respondent draw weights

$$\begin{aligned} W_{gi}^{(\mathrm{R})} &= \frac{m_g A_g B_{gi}}{\sum_h m_h A_h} \quad \text{(respondent target)}, \\ W_{gi}^{(\mathrm{C})} &= A_g B_{gi} \quad \text{(equal-cluster target)}. \end{aligned} \tag{4}$$

Each draw evaluates H on the reweighted distribution $F^{(W)} = \sum_{g,i} W_{gi}\, \delta_{Y_{gi}}$ via the exact transport. The law satisfies exact identities that fix its semantics, each enforced by the verification suite: weights are nonnegative and sum exactly to one; replacing all Dirichlet draws by their means returns exactly the target's point weighting ($1/N$ per respondent for F_{R}; $1/(Gm_g)$ for F_{C}), so the two targets are algebraically separated draw by draw; and the law is invariant to cluster relabeling and to within-cluster row relabeling, with unequal sizes and single-respondent clusters handled as exact special cases. The raw interval takes B draws and reports the type-8 equal-tail 2.5%/97.5% quantiles of the defined draws $[L_n, U_n]$, at least 99% of draws being required to be defined.

We now consider why the raw interval might be expected to miscover, and in which direction; a heuristic account suggests both. The independent sampling unit is the cluster, so a correctly scaled interval should shrink at the $\sqrt{G}$ rate. The two-stage law, however, compounds *two* independent Dirichlet layers, and at finite G the within-cluster layer contributes weight variability that the sampling process (which fixes each sampled cluster's respondents) need not reproduce, plausibly inflating the draw distribution. We emphasize that this is a candidate mechanism, not a theorem; what is established is empirical. Our pilot established the direction and magnitude: the raw two-stage interval covered .992–.997 across primary-target lanes, and its diagnostic decompositions are consistent with the two-layer reading from both sides — the within-cluster-only law (cluster weights fixed) covered .86–.90, and the one-stage law (cluster weights only, within-cluster distributions fixed) covered .90–.93, each alone anti-conservative while the combination overcovers. Overcoverage is not innocuous, since the excess width understates the precision that the data support. We treat this as a finite-G width problem and correct it by the smallest deterministic modification available rather than by re-engineering the law itself.

3.4. **The Cluster-Rate Width Calibration.** Let $[L_n, U_n]$ be a defined raw interval with midpoint m_n and half-width w_n. The calibrated interval is

$$[\widetilde{L}_n, \widetilde{U}_n] \;=\; \bigl[m_n - k_G w_n,\; m_n + k_G w_n\bigr], \qquad k_G \;=\; 1 + \frac{\gamma}{\sqrt{G}}, \qquad \gamma = -1.5, \tag{5}$$

with no endpoint clipping, inherited typed failure status, and no per-replication method switching; the multiplier is required to be positive in every evaluated design ($k_G = .68$ at $G = 22$, $.726$ at the SWMDK $G = 30$, $.83$ at $G = 80$). Three properties are exact for any $\gamma \in (-\sqrt{G}, 0]$: the midpoint is unchanged; the length is multiplied by exactly k_G; and, in the reverse of the widening case, $[\widetilde{L}_n, \widetilde{U}_n] \subseteq [L_n, U_n]$, so that on any replication set the calibrated interval's coverage is at most the raw interval's. Shrinkage can therefore only reduce coverage; its justification is that the raw interval demonstrably covers at a higher rate than it states, and the frozen gates make the consequences of the reduction auditable. For fixed γ, the endpoint perturbation is $\mp\gamma w_n/\sqrt{G}$; if the raw half-width is $O_p(G^{-1/2})$ — the cluster-rate behavior the design targets — the perturbation is $O_p(G^{-1})$, and by Slutsky's theorem the calibrated endpoints share whatever $\sqrt{G}$-limit the raw endpoints possess, leaving first-order coverage unchanged in the limit (proofs in Appendix B).

One boundary should be drawn sharply: we do not claim a conditional limit theorem establishing that the two-stage law matches the first-order sampling law of the clustered functional, in the way such a theorem is available for the independent-respondent BB (Lo, 1987). The hierarchical law's justification in this article is architectural (target-exact weight algebra at the cluster rate), diagnostic (the overcoverage decomposition above, read as evidence consistent with the two-layer account rather than proof of it), and, most importantly, empirical: a locked train–validate–confirm sequence whose gates were recorded before the data that judged them. The constant $\gamma = -1.5$ is simulation-calibrated, not analytically derived, and its selection chronology is described in the next section. It is also not portable: the independent-respondent binary calibration of Lee (2026b) requires a *positive* constant at the $\sqrt{n}$ rate, and the frozen protocol forbids transferring either constant to the other setting.

4. The Calibrated Interval Procedure

4.1. **Exact Algorithm.** For a fixed polytomous item set with declared score supports, complete responses from N respondents in G clusters, level 95%, and B weight draws:

(1) *Weights.* Draw B independent two-stage weight vectors from Equation (3), combined per Equation (4) for the declared target (respondent-weighted or equal-cluster), from a seeded, draw-major stream whose output is identical for any batch size and worker count.

(2) *Functional.* For each draw, evaluate $H(F^{(W)})$ of Equation (2) exactly: weighted score margins and cross-moments per pair, denominator by the two-pointer comonotone transport. Draws with a nonpositive weighted scale denominator are typed invalid, not repaired.

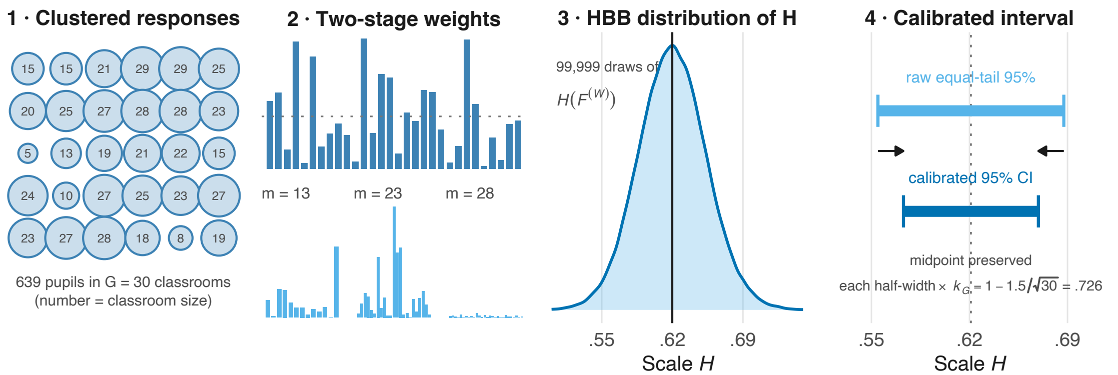
(3) *Raw interval.* If at least 99% of draws (and at least the frozen minimum count) are defined, $[L_n, U_n]$ is the type-8 equal-tail quantile pair of the defined draws.

(4) *Calibration.* Report $[m_n - k_G w_n,\ m_n + k_G w_n]$ with $k_G = 1 - 1.5/\sqrt{G}$, per Equation (5).

Figure 1 shows the four steps end to end on the school data of Section 6: the nested sample, one two-stage weight draw, the draw distribution of the coefficient, and the shrinkage of the raw interval about its fixed midpoint.

Failure semantics are strict and inherited: an undefined draw or interval remains undefined, an undefined interval counts as a miss in every coverage denominator, and neither endpoint clipping nor per-replication cascading is permitted; Appendix C records the locked method identifiers and the typed failure taxonomy. The level is

FIGURE 1. The Calibrated Hierarchical Bayesian Bootstrap Interval, End to End



Note. Illustration with the SWMDK application (well-being with teachers, respondent-weighted target). (1) 639 pupils are nested in $G = 30$ classrooms of 5 to 29 pupils. (2) One illustrative two-stage weight draw (display seed): classroom weights $A_g \sim \text{Dirichlet}_G(1, \dots, 1)$ (top; dotted line at $1/G$), then within-classroom weights B_{gi} combine into respondent draw weights $W_{gi} \propto m_g A_g B_{gi}$ (bottom; three classrooms shown). (3) The 99,999 locked production draws of the weighted polytomous coefficient H under the hierarchical law (vertical line: the sample estimate). (4) The raw equal-tail 95% interval *shrinks* about its fixed midpoint by $k_G = 1 - 1.5/\sqrt{30} = .726$, giving the calibrated interval. BB = Bayesian bootstrap; CI = confidence interval.

fixed at 95%, the only level with calibration evidence, and γ is not a user parameter. The calibration evidence spans 22 to 80 clusters, and the shipped implementation enforces that envelope (Section 4.3). Each stage ran at its own protocol-fixed draw budget: the development and held-out validation simulations used $B = 299$ draws per replication, the confirmation $B = 499$, and the empirical application $B = 99{,}999$ per task.

4.2. **Selection of the Constant $\gamma = -1.5$.** The chronology ran in one direction, with each stage completed and locked before the next produced outcomes (Figure 2c); Appendix E reproduces the stage-by-stage decisions and gate results.

Pilot. A feasibility pilot (1,080 replications across regular and knot lanes) established the phenomenon: the raw two-stage interval covered .992–.997 across the primary (respondent) target's lanes (.989–1.000 over all target–lane combinations), the working-independence and one-stage intervals undercovered, and the two-level delta and frequentist cluster bootstrap intervals were above nominal. The pilot established the diagnosis (a finite-G width problem) and nothing else; no calibration yet existed.

Training. The candidate family was frozen as the midpoint-preserving multiplier $k_G = 1 + \gamma/\sqrt{G}$ on the grid $\gamma \in \{0, -.5, -.75, -1, -1.25, -1.5\}$, applied to both targets. On 24 training cells (120 replications each; 2,880 scheduled), frozen eligibility screens (coverage bands [.94, .98] for the respondent target and [.94, .985] for the equal-cluster target, subgroup floors by G and by cluster-size mechanism, an invalid-fraction cap, and multiplier positivity in every cell) eliminated $\gamma \in \{0, -.5, -.75, -1\}$, in every case for residual overcoverage. Exactly two candidates survived, $\gamma = -1.25$ and $\gamma = -1.5$, and the prespecified rule (minimum mean interval length among eligible candidates) selected -1.5, whose primary-target intervals were 6.3% shorter; neither tie-break rule was needed. The lock was serialized before any validation outcome existed.

Held-out validation. On 18 validation cells disjoint from training in both conditions and seeds (180 replications each; 3,240 scheduled), the locked candidate covered .9718 (respondent) and .9722 (equal-cluster), inside its frozen acceptance bands, with zero invalid replications. The prespecified fallback (reverting to the raw interval if the candidate failed) was itself evaluated and failed its own bands (.9949 and .9968, above the caps), so the fallback was excluded on the basis of evidence rather than discretion. The final method lock (`LOCKED_FOR_PHASE16_CONFIRMATION`) froze the procedure, identifiers, and confirmation gates before any confirmatory data were generated; nothing changed afterward.

The status of the constant follows from this chronology: $\gamma = -1.5$ is a simulation-calibrated cluster-rate quantity, selected in development, validated on disjoint held-out conditions, and prospectively confirmed. It is not analytically derived, and it is not transferable to the independent-respondent setting of Lee (2026b).

4.3. **Software.** The procedure ships in the open-source R package bayesmokken (Lee, 2026a), through the clustered-ordinal path of the public entry point:

```
fit <- bb_mokken_h(data, item_type = "ordinal", supports = supports,
                   cluster = class_id,
                   cluster_target = "respondent",      # or "equal_cluster"
                   weight_law = "hierarchical_cluster",
                   draws = 2000, seed = 1, interval = "calibrated")
fit$interval
```

The interface is estimand-first by construction: the population target and the item score supports must be declared rather than defaulted, so no weighting convention can select a population coefficient silently, and the fitted object records its target, weight law, and method identifier so that an archived result is self-describing. Requests outside this article's evidence are refused rather than reinterpreted: binary items with a cluster argument, ordinal items under the independent-respondent law, cluster-free ordinal random-weight intervals, and calibrated intervals for fewer than 22 clusters are validation errors, and fits beyond 80 clusters are labeled as extrapolated. The same package implements, on a separately validated path, the independent-respondent binary method of Lee (2026b); the package enforces the boundary between the two procedures, and none of the present evidence bears on that path. Package-level verification evidence, the returned object's contract, and the release history are collected in Appendix H.

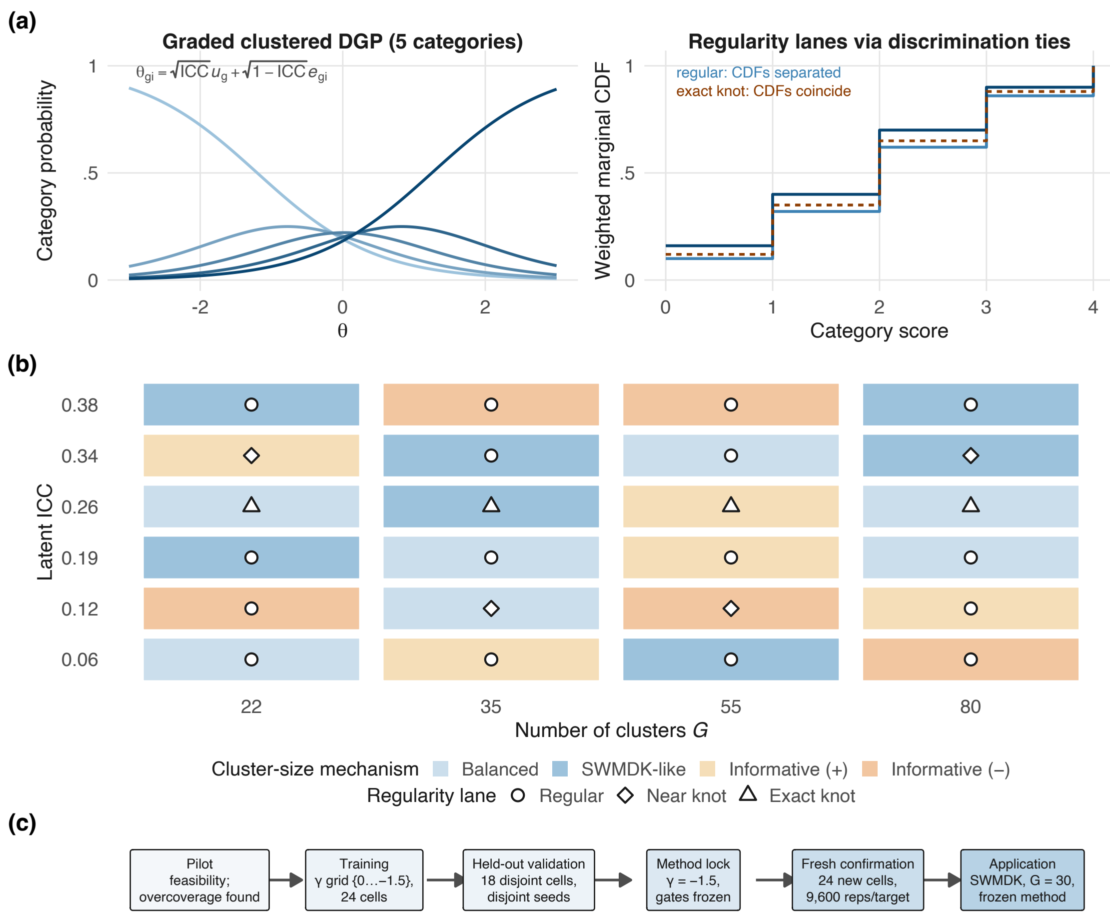
4.4. **Transparency and Openness.** We report how every design condition, Monte Carlo budget, and analysis rule was determined; there were no data exclusions, and no analyses beyond the frozen protocols were run. Simulation reporting follows Morris et al. (2019), with Monte Carlo standard errors attached to every coverage estimate (Koehler et al., 2009). The study was not preregistered in an external registry; instead, every stage, from the estimand and sampling contract to the application protocol, was serialized before its outcomes existed into hash-locked protocol and receipt files that are archived, with their SHA-256 digests and the scripts that verify them, in the replication package (Appendix E). A frozen claim-eligibility register written at the source-review gate governs the wording of this article and is reproduced in Appendix H. The empirical data are public: `SWMDK` is distributed with the mokken package (van der Ark, 2007), and our pipeline consumed it read-only with a locked digest; Appendix G records the provenance chain in full. Analyses used R version 4.6.0 (R Core Team, 2026). All code, locked artifacts, and the figure/table pipeline are available in the replication archive at https://github.com/joonho112/bayesmokken-clustered-polytomous-replication, and the method is implemented in bayesmokken at https://github.com/joonho112/bayesmokken (Lee, 2026a).

5. Confirmatory Simulation

The confirmation evaluates the single frozen procedure ($\gamma = -1.5$, both targets, gates included) on conditions generated after the lock and never touched during development.

5.1. **Design.** *Data-generating process.* Respondents answer $J = 5$ five-category items (0–4) under a graded cumulative-logit model (Samejima, 1969) with conditional independence given $\theta_{gi} = \sqrt{\mathrm{ICC}}\, u_g + \sqrt{1 - \mathrm{ICC}}\, e_{gi}$; three fresh threshold profiles frozen

FIGURE 2. The Evaluation Design at a Glance



Note. (a) The clustered graded data-generating process (illustrative parameters). Left: a five-category cumulative-logit item driven by $\theta_{gi} = \sqrt{\text{ICC}}\, u_g + \sqrt{1 - \text{ICC}}\, e_{gi}$ with $u_g, e_{gi} \sim \mathcal{N}(0, 1)$. Right: the regularity lanes — distinct item discriminations keep weighted marginal CDFs separated (regular); equating discriminations makes cumulative margins coincide (exact knot), the configuration in which the transport path can switch. (b) The 24 fresh confirmation cells: number of clusters G by latent ICC, with fill showing the cluster-size mechanism and symbols the regularity lane. (c) The frozen chronology; no stage was revisited after its outcomes existed. Reps = scheduled replications; SWMDK = the empirical application data.

for this stage (standard, rare-high, and rare-low *confirm* variants, disjoint from the six used in development and validation) and item-specific location offsets and discrimination multipliers complete the item side (Figure 2a). Population truth for both targets is computed by two-dimensional Gauss–Hermite quadrature (51 nodes per dimension under the confirmation protocol; development and validation used 41) with target-specific cluster-size weighting, and never by simulation; Appendix D specifies the mechanism, the truth oracle, and all three designs in full.

Cells and lanes. The 24 fresh cells (Figure 2b) cross the number of clusters $G \in \{22, 35, 55, 80\}$ with latent ICC from .06 to .38, four cluster-size mechanisms (balanced with $m_g = 20$; SWMDK-like, resampled from the observed 5–29 sizes; and informative sizes in both directions, where m_g increases or decreases with the cluster effect u_g, making the respondent-weighted and equal-cluster targets genuinely diverge), and fresh threshold/discrimination parameterizations. Sixteen cells are regular (the decision lane); four near-knot and four exact-knot cells stress the transport path and are diagnostic by contract. Each cell ran 400 replications with $B = 499$ draws per replication (the confirmation protocol's own budget, raised from the 299 used in development) and per-target truth: 9,600 scheduled replications per target for the selected method, 6,400 per target for the raw comparator on the 16-cell decision lane, and full comparator batteries on all three lanes.

Comparators. Five frozen alternatives run on identical replications: the naive iid respondent BB (a negative control targeting the working-independence functional; its use on clustered data is exactly the practice the article warns against); a one-stage cluster BB (cluster weights only, within-cluster distributions fixed); a two-stage frequentist cluster bootstrap (resample G clusters, then respondents within each); the two-level delta-method Wald interval of Koopman, Zijlstra, and van der Ark (2020); and the within-cluster-only HBB (diagnostic decomposition). Exact specifications, including typed failure conditions, are in Appendix C.

Gates. Seven blocking criteria were frozen with the method: family-level unconditional coverage inside [.935, .975] (respondent) and [.93, .98] (equal-cluster) on the regular lane; a regular-cell floor of .88; subgroup floors of .91 by G and by size mechanism; an invalid-fraction cap; and a length-efficiency requirement that the mean calibrated length be at most 90% of raw. Coverage denominators are all scheduled replications.

5.2. **Results.**

5.2.1. *Primary Coverage and the Overcoverage Repair.* All seven gates passed. Figure 3 and Table 1 give the principal results: on the regular lane the calibrated interval covered .9736 (respondent; MCSE .0020) and .9738 (equal-cluster; .0020), inside both frozen bands. The respondent-target estimate sits within one Monte Carlo standard error of its band ceiling of .975: the locked shrinkage passes at the conservative edge of its own specification, a point the subgroup analysis below localizes and the Discussion's first future direction takes up. The raw two-stage interval covered .9975 (respondent) and .9962 (equal-cluster), confirming on untouched conditions that the raw law overcovers. The cost of the repair is modest: the calibrated interval's mean length is 75.1% of raw (the gate required at most 90%), and there were no invalid replications anywhere in the design. Cell-level coverage for the selected method spans .925–.995 across all 24 cells and both targets (regular-lane minimum .930 against the .88 floor; the all-lane minimum, .925, occurs in a near-knot diagnostic cell), and no interval in any cell of any method produced an invalid replication; Appendix F reports every cell.

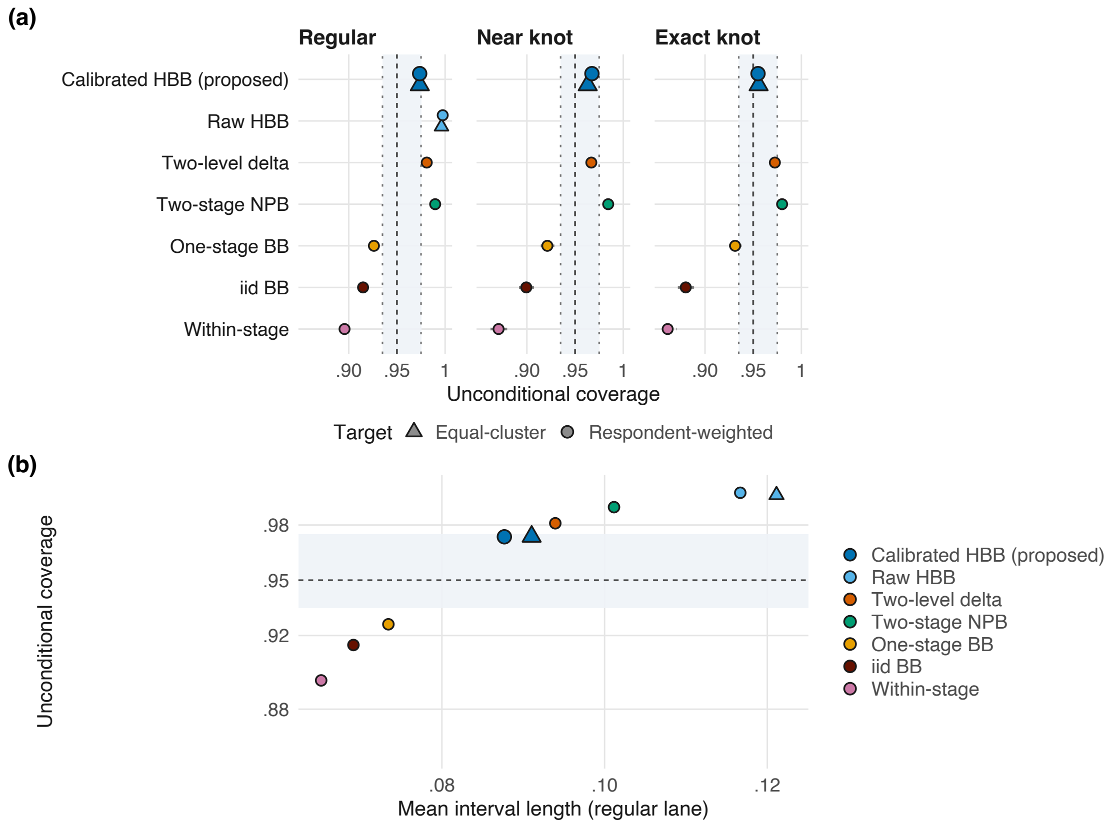
Table 1. Fresh-Confirmation Coverage and Length by Regularity Lane

	Regular lane (decision)		Near knot	Exact knot
Interval procedure	Coverage (MCSE)	Length	Coverage (MCSE)	Coverage (MCSE)
Calibrated HBB — respondent target	.9736 (.0020)	0.0877	.9675 (.0044)	.9550 (.0052)
Calibrated HBB — equal-cluster target	.9738 (.0020)	0.0910	.9631 (.0047)	.9556 (.0051)
Raw HBB ($\gamma = 0$) — respondent	.9975 (.0006)	0.1167	—	—
Raw HBB ($\gamma = 0$) — equal-cluster	.9962 (.0008)	0.1211	—	—
Two-level delta (analytic)	.9809 (.0017)	0.0939	.9669 (.0045)	.9725 (.0041)
Two-stage cluster NPB	.9897 (.0013)	0.1012	.9844 (.0031)	.9800 (.0035)
One-stage cluster BB	.9261 (.0033)	0.0734	.9213 (.0067)	.9313 (.0063)
iid respondent BB (naive)	.9148 (.0035)	0.0691	.8994 (.0075)	.8800 (.0081)
Within-cluster HBB (diagnostic)	.8956 (.0038)	0.0652	.8706 (.0084)	.8612 (.0086)

Note. Fresh, untouched confirmation: 16 regular cells (6,400 scheduled replications per target), 4 near-knot and 4 exact-knot diagnostic cells (1,600 each), 400 replications per cell, 499 weight draws per replication. Coverage is unconditional; no interval in the table produced an invalid replication (maximum invalid fraction 0). The frozen decision bands apply to the regular lane only: [.935, .975] for the respondent target and [.93, .98] for the equal-cluster target. The raw hierarchical BB was evaluated on the decision lane; comparators (respondent target; the iid BB addresses the working-independence respondent target) were evaluated on all three lanes. HBB = hierarchical Bayesian bootstrap; NPB = nonparametric bootstrap; MCSE = Monte Carlo standard error.

5.2.2. *Comparator Failure Modes.* The comparators fail in different and informative directions (Figure 3b). The naive iid BB covers .915 on the regular lane, and as little as .880 in the exact-knot lane, at lengths roughly 21% shorter than is warranted; ignoring clustering therefore not only misstates precision but overstates it systematically. The one-stage cluster BB (.926) shows that weighting clusters alone is not sufficient, and the within-cluster-only decomposition (.896) shows the complementary failure; together they isolate the two components that the two-stage law combines. On the conservative side, the two-stage frequentist bootstrap covers .990 at the longest

FIGURE 3. Fresh-Confirmation Coverage and the Length–Coverage Frontier



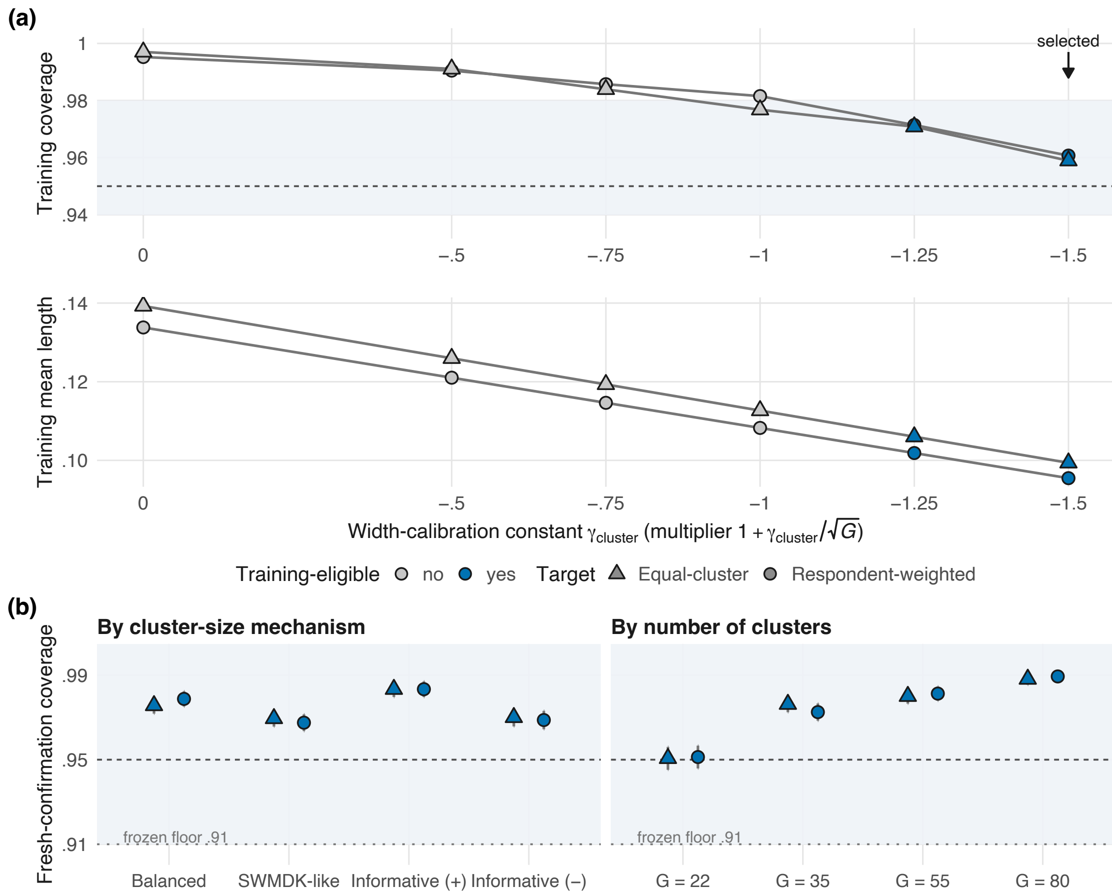
Note. (a) Unconditional coverage by regularity lane. The calibrated HBB (dark blue, enlarged) is shown for both targets (circles: respondent; triangles: equal-cluster); the raw HBB was evaluated on the regular decision lane; comparators (respondent target; the iid BB addresses the working-independence target) run on all lanes. Shading: the frozen respondent-target decision band [.935, .975]; dashed line: nominal .95. Error bars are ± 1 MCSE (mostly smaller than the symbols). (b) The regular-lane frontier: mean interval length versus coverage. Methods below the band are anti-conservative at cluster-blind lengths; methods above it buy their coverage with width. HBB = hierarchical Bayesian bootstrap; NPB = nonparametric bootstrap.

lengths in the design (15% longer than the calibrated interval), and the two-level delta interval, the strongest available practice baseline, covers .981 at 7.1% greater length than ours. We report these as design-specific comparisons under the frozen protocol rather than as universal orderings; what they establish is that, within this design, the calibrated HBB is the only procedure that falls inside the decision bands, and that it does so at the short end of the admissible-length range.

5.2.3. *Calibration Anatomy and Subgroup Behavior.* Figure 4 examines the constant in more detail. Panel (a) shows the training dose–response: coverage and length fall monotonically as γ moves from 0 to -1.5, the first four grid points fail the frozen eligibility screens for residual overcoverage, and of the two surviving candidates the shorter is selected. Panel (b) shows the fresh-confirmation subgroups that the gates protect: coverage by number of clusters runs from .951 at $G = 22$ to .989 at $G = 80$ (floor .91), and by size mechanism from .9675 upward, including both informative mechanisms, where target separation matters most. The pattern at large G is residual conservatism rather than undercoverage: a single constant cannot be exactly nominal at every G, and the frozen bands allowed for this. The knot lanes behave as intended: the selected interval's diagnostic coverage is .955–.968, degraded relative to the regular lane but well away from the deterioration that the naive comparators show there; by contract these lanes support disclosure rather than primary claims.

5.2.4. *Monte Carlo Precision and Independent Checks.* Regular-lane MCSEs are .0020 at the lane level and about .011 at the cell level (400 replications); every reported contrast between the selected and raw intervals is a paired comparison on identical replications and draws. The confirmation decision (`CONFIRM_CLUSTER_POLYTOMOUS_METHOD`, 7 of 7 checks) was issued by the frozen evaluator on raw outputs downloaded in full before any evaluation, and the subsequent source-review and production gates re-verified the chain (13 of 13 and 19 of 19 criteria, respectively) without touching any scientific number. These results pertain to the frozen design; the limitations subsection of the Discussion collects what remains open.

Figure 4. Anatomy of the Cluster-Rate Calibration



Note. (a) Training-stage dose–response of the width constant: unconditional coverage (top; shading = frozen respondent eligibility band $[.94, .98]$; dashed = nominal) and mean length (bottom) across the frozen grid, by target. Gray points failed at least one frozen eligibility screen; blue points were eligible; the arrow marks the selected $\gamma = -1.5$ (shortest eligible). (b) Fresh-confirmation coverage of the locked method by number of clusters and by cluster-size mechanism, with ± 1 MCSE bars and the frozen subgroup floor .91 (dotted).

6. Prespecified Application: Well-Being in Classrooms (SWMDK)

6.1. **Data and Prespecification.** The SWMDK data (distributed with mokken; Koopman et al., 2022; van der Ark, 2007) are a documented subset of the Dutch COOL5-18 cohort study (Zijsling et al., 2017a, 2017b): 639 secondary-school pupils nested in $G = 30$ classrooms of 5 to 29 pupils (median 23), answering five-category items from the school-well-being scales of Peetsma et al. (2001). The two analysis scales are the final scales of the two-step, test-guided analysis that Koopman et al. (2022) performed on these data: well-being with teachers (Items 1–6) and well-being with classmates (Items 8, 9, 10, 11, and 13), from which their procedure had removed Items 7 and 12 as contributing too little to accurate measurement. We adopted those scales as fixed, performed no item selection, deletion, or re-keying of our own, and designated threshold-based scale classification as out of scope by protocol.

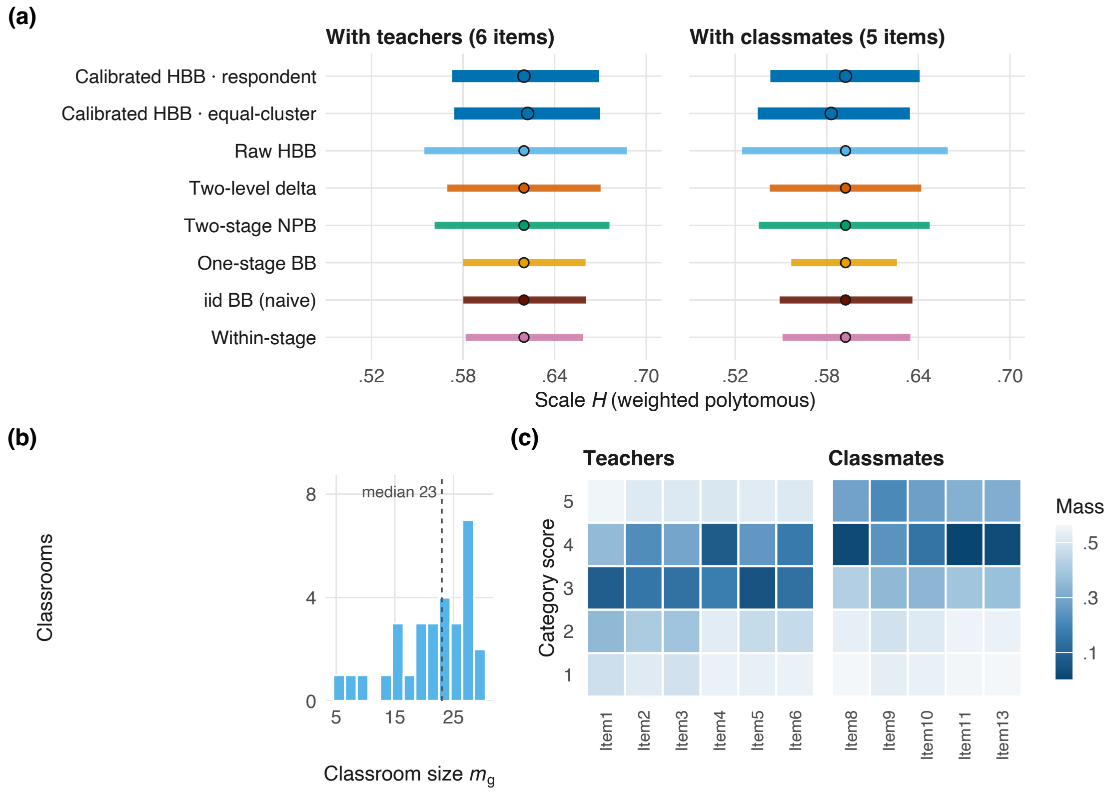
A low-precision feasibility prototype (1,999 draws per interval), run during early scoping, had already displayed the four point estimates, so the outcome-blindness claim made here is narrower than complete ignorance of the results. Before any of the twelve high-precision production tasks ran, the analysis contract froze the item sets, the classroom identifier as the cluster, both targets, the selected method and its locked γ, the full comparator battery, the seeds, the precision and split-half thresholds, and the evaluator, with production output counts of zero at authority, protocol-lock, and code-lock time, and with nothing chosen or altered in response to the prototype. Every fixed item retains all five categories in the data (smallest category mass .0031), and clustering is clearly nonignorable: intraclass correlations are .169 (teacher) and .183 (classmate), the values Koopman et al. (2022) also report. Appendix G gives the provenance, the full category profile, and the stability diagnostics.

The application ran as twelve computation tasks with 99,999 weight draws each (1,199,988 draws in total, none invalid), and the four selected intervals are deterministic k_G transforms of their raw draw streams with $k_G = 1 - 1.5/\sqrt{30} = .7261$. All 14 blocking application-gate criteria passed, including point-estimate concordance with an independent implementation below 10^{-10} and split-half endpoint stability within .0009 against a frozen .005 threshold.

6.2. **Results.** Table 2 and Figure 5 present the estimates. For well-being with teachers, the respondent-weighted coefficient is $\widehat{H} = .6199$ with calibrated 95% CI $[.5728, .6692]$; for well-being with classmates, $\widehat{H} = .5923$, CI $[.5431, .6408]$. Under the equal-cluster target the point estimates move to .6223 and .5829 (differences of $+.0024$ and $-.0094$), so the two estimands, which coincide only under noninformative size, here lead to compatible conclusions while remaining visibly distinct quantities; both are reported and labeled throughout. Descriptively, both scales sit in territory conventionally described as strong scalability; consistent with the frozen protocol, we attach no formal threshold classification to that reading.

The comparator pattern mirrors the simulation results. The calibrated intervals are 73% of the raw HBB's width on both scales (the same deterministic ratio that the confirmation validated) and lie just inside the two-level delta intervals (96.0%

FIGURE 5. SWMDK: Intervals, Clustering, and Category Structure



Note. (a) 95% intervals for both fixed scales: the calibrated HBB under both targets (dark blue, enlarged), the raw HBB, and the frozen comparators (respondent target). No threshold reference lines are drawn; the frozen protocol excludes threshold-based classification. (b) Observed classroom sizes. (c) Category masses of the fixed items; every item retains all five categories. HBB = hierarchical Bayesian bootstrap; NPB = nonparametric bootstrap.

and 98.5% of their lengths for teachers and classmates); that two different inferential routes, one analytic and one based on reweighting, arrive at nearly the same uncertainty is reassuring, and it is not presented as a superiority claim. On the anti-conservative side, the naive iid interval is 17%–11% shorter than the calibrated one (width that the simulation indicates is not supported at these intraclass correlations), and the one-stage and within-stage intervals are shorter still.

One diagnostic finding should also be reported. Under the respondent-weighted target, both scales' weighted marginal configurations lie exactly at cumulative knots (two per scale, with two near-knots each; minimum internal CDF gap 0), whereas under the equal-cluster target the same data produce no knots (minimum gaps .0012 and .0006). The target weighting, rather than the responses, changes the tie structure

TABLE 2. SWMDK Application: Selected Intervals, Targets, and Reference Comparators

	Well-being with teachers		Well-being with classmates	
	Respondent	Equal-cluster	Respondent	Equal-cluster
Point estimate $\widehat{H}$	.6199	.6223	.5923	.5829
Calibrated 95% CI	[.5728, .6692]	[.5743, .6698]	[.5431, .6408]	[.5347, .6345]
Interval length	.0964	.0956	.0978	.0998
Raw HBB 95% interval	[.5547, .6874]		[.5246, .6593]	
Two-level delta 95% interval	[.5697, .6701]		[.5427, .6419]	
Selected / raw length	.7261		.7261	
Selected / two-level delta length	.9599		.9854	
Intraclass correlation	.169		.183	
Exact cumulative knots (resp.; eq.-cl.)	2; 0		2; 0	

Note. Fixed prespecified scales (teacher: Items 1–6; classmate: Items 8, 9, 10, 11, 13); 639 pupils in $G = 30$ classrooms; 99,999 weight draws per task with zero invalid draws; the calibrated interval multiplies each raw half-width by $k_G = 1 - 1.5/\sqrt{30} = .7261$ about a fixed midpoint. Raw HBB and two-level delta rows are respondent-target reference comparators. Exact-knot counts are diagnostic disclosures of the weighted-marginal transport path (Appendix G), not error flags. No threshold-based scale classification is performed. CI = confidence interval; HBB = hierarchical Bayesian bootstrap.

of the weighted CDFs, which illustrates with real data why the knot lanes exist in the design. Under the frozen contract this is a diagnostic statement about transport-path geometry rather than a caveat on the reported coefficients, which remain exactly computed in either case; the simulation's exact-knot lane (.955 coverage for the selected method) indicates the extent of the associated cost.

7. Discussion

7.1. **Summary of Findings.** For fixed polytomous scales administered in clusters, this article provides an uncertainty statement that specifies which population coefficient it concerns and that holds its advertised coverage under prespecified, demanding evaluation. The components are separable and were validated separately. The estimand layer distinguishes respondent-weighted from equal-cluster functionals and carries both through every table. The computation layer evaluates Molenaar's weighted coefficient exactly, under arbitrary weights, by finite comonotone transport, implementing established results (Molenaar, 1991; Rüschendorf, 1982), with software precedent credited (Andreadis, 2017) and with oracle-verified code rather than new theorems. The inference layer pairs a target-exact two-stage hierarchical weight law with a cluster-rate width calibration whose constant was fixed on training data, was validated on disjoint held-out conditions while its prespecified fallback failed, and then held its frozen bands on the fresh confirmation's 16-cell regular decision lane, with .974 coverage for both targets at three-quarters of the raw width, every gate passing, and all 24 untouched cells, diagnostic knot lanes included, disclosed cell by cell. The application layer shows the frozen procedure producing stable, cluster-aware intervals on real school data at 96.0% and 98.5% of the analytic two-level interval's width for the two scales, with full diagnostic disclosure.

The comparative pattern carries the main practical implication. Ignoring clustering (the naive iid BB) undercovers at .88–.92 while appearing 11%–21% more precise than is justified; partial corrections (one-stage and within-stage laws) remain anti-conservative; the frequentist two-stage bootstrap and the raw HBB overcover at a corresponding cost in width; and the analytic two-level delta interval is a genuinely strong baseline that this article complements with target explicitness, finite-G calibration evidence, and derivative-free mechanics, at modestly shorter length in every matched comparison we ran. None of this constitutes a universal ordering; all of it is a locked-design finding whose transportability the gates were constructed to examine, subgroup by subgroup.

7.2. **Contrast With the Independent-Respondent Calibration.** Read together with Lee (2026b), this article suggests caution in treating bootstrap calibration constants as portable. For binary scales with independent respondents, that article's raw respondent-level BB interval undercovers slightly and its validated correction widens it at the $\sqrt{n}$ rate. Here, for polytomous scales with clustered respondents, the raw two-stage law overcovers substantially and the validated correction shrinks the interval at the $\sqrt{G}$ rate: the sign is opposite, the rate variable differs, and so does the magnitude. The direction of a weight law's finite-sample miscalibration evidently depends jointly on the functional, the weighting architecture, and the sampling structure, and in our view it must be established by governed experimentation rather than assumed. Our protocols encode this as a firewall: neither constant may be transferred to the other setting, and the package exposes neither as a tuning parameter.

7.3. **Practical Guidance.** Three recommendations for applied two-level Mokken analysis follow directly. First, when respondents are nested and the intraclass correlation is nonnegligible, an independence-based interval for H should not be reported; at the intraclass correlations of ordinary classroom data the coverage shortfall is large and structural, and it cannot be detected from the interval itself. Second, the target should be declared: with unequal cluster sizes, it should be stated whether the claim concerns the average respondent or the average cluster, and if size could be informative both targets should be reported. In SWMDK the two moved in opposite directions across scales while supporting the same substantive conclusion, which is precisely the kind of robustness information a reader should receive. Third, the conventional H benchmarks should be treated as descriptive vocabulary rather than as decision thresholds; this article performs no threshold classification, and its intervals are not significance tests against .5. The one-call bayesmokken interface makes the cluster-aware analysis no more burdensome than the independence-based one.

7.4. **Scope and Limitations.** The frozen claim register (Appendix H) governs this article, and its prohibitions are worth restating in prose. We claim no priority for weighted Mokken computation, since Andreadis (2017) precedes us, and no new transport theorem: the denominator mathematics is Rüschendorf's and Molenaar's, implemented and specialized. We claim no universal superiority over any comparator; the evidence is bounded by the frozen designs, and the two-level delta interval in particular remains an excellent instrument whose differences from ours concern semantics (targets, symmetry, derivatives) as much as width. We make no threshold-based scale-quality decision for SWMDK. We do not treat exact-knot configurations as regular inference territory; they are exactly computable and fully disclosed, but the associated theory is unfinished. We prove no conditional limit theorem for the hierarchical law; its warrant here is architectural, diagnostic (the overcoverage decomposition is evidence consistent with the two-layer account, not proof of it), and confirmatory-empirical. Finally, the intervals are BB-based confidence intervals under repeated sampling of clusters, and at no point posterior credible intervals.

7.5. **Future Directions.** Four lines of work appear most productive. First, limit theory: a conditional central limit theorem for two-stage Dirichlet functionals of kinked ratios, and ideally a higher-order account of the raw law's overcoverage, would convert the calibration from validated practice into derived correction and might yield a G-dependent replacement for the constant, whose finite-sample cost our results document as mild large-G conservatism and a respondent-lane coverage estimate at the top of its frozen band. Second, nonregular theory at knots: directional asymptotics for the transport path would upgrade the diagnostic lanes into inference lanes. Third, design extensions in order of applied urgency: missing responses, three-level nesting, small-G regimes below 22, and item-level coefficients with cluster-aware uncertainty. Fourth, unification: a framework predicting the sign and size of weight-law miscalibration across sampling structures would subsume both this article's shrinkage and the widening of Lee (2026b) as special cases, and is in our view the most interesting open problem that the pair of studies raises.

7.6. **Concluding Remarks.** Every constant, gate, comparator, and seed in this article was committed to disk, hashed, and sealed before the data that judged it existed, and the claim register that disciplines this text was frozen at the source gate (for the surrounding transparency literature, see Depaoli & van de Schoot, 2017; Nosek et al., 2015; Wilson et al., 2014). The result is a cluster-aware uncertainty statement for a widely used measurement coefficient that was evaluated under its own prespecified rules, and we hope that the same approach, in which bootstrap calibrations are constructed, audited, and bounded in the manner of confirmatory research, will prove useful more generally.

Acknowledgments. The empirical illustration was made possible by the COOL5-18 cohort study and by the distribution of the `SWMDK` subset with the mokken package; the author thanks the authors of both for that openness, and the two-level Mokken literature on which this article builds for documenting the workflow and the subset construction it adopts.

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Software and reproducibility. The analyses used R 4.6.0 (R Core Team, 2026); the calibrated cluster-aware interval is implemented in the R package bayesmokken (Lee, 2026a), and point-estimate concordance was checked against an independent implementation. Every stage of the study (the estimand and sampling contract, the weight-law specification, the candidate grid, eligibility and acceptance bands, the confirmation design and gates, and the application protocol) was serialized into hash-locked protocol and receipt files before the outcomes it governs were generated; those files, their SHA-256 digests, the complete raw draw streams of the application, and the scripts that verify them ship with the replication archive. The package is available at https://github.com/joonho112/bayesmokken and the replication archive at https://github.com/joonho112/bayesmokken-clustered-polytomous-replication.

Data availability. The empirical data are public: `SWMDK` is distributed with the mokken package (van der Ark, 2007) as a documented subset of the Dutch COOL5-18 cohort study (Zijsling et al., 2017a, 2017b), with items from Peetsma et al. (2001) and subset construction per Koopman et al. (2022). The analysis pipeline consumed it read-only under a locked digest.

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Online Supplement

Cluster-Aware Uncertainty for Polytomous Mokken Scalability:
A Calibrated Hierarchical Bayesian Bootstrap

JoonHo Lee

This supplement accompanies the main text. Section, equation, table, and figure numbers below are prefixed by the appendix letter (A.1, B.1, ...). References to the main text use its unprefixed numbering.

Contents

Appendix A. Estimands, Notation, and Regularity

A.1. **Sampling Frame and Targets.** Clusters $g = 1, \dots, G$ are independent draws of (m_g, F_g) from a superpopulation: a cluster size and a within-cluster response distribution on the fixed item set, possibly dependent (informative size). Respondents are exchangeable within clusters given (m_g, F_g), with $N = \sum_g m_g$. Writing (M, F) for one superpopulation draw, the two population response distributions of Equation (1) in the main text, restated here as Equation (A.1),

$$F_{\mathrm{R}} = \frac{\mathbb{E}(MF)}{\mathbb{E}(M)}, \qquad F_{\mathrm{C}} = \mathbb{E}(F), \tag{A.1}$$

induce the two scalability estimands $H(F_{\mathrm{R}})$ (primary, respondent-weighted; the size-biased cluster mixture) and $H(F_{\mathrm{C}})$ (sensitivity, equal-cluster). These are population functionals of the superpopulation law, not expectations of finite-G sample ratios; the truth oracle evaluates them by size-weighted (respectively unweighted) quadrature over the cluster-effect distribution (Appendix D). Under noninformative size ($m_g \perp\!\!\!\perp F_g$) they coincide; under informative size they need not, and the informative cells of the simulation are built so that they measurably differ. The finite-sample plug-ins reweight the sampled clusters accordingly — $\sum_g m_g F_g / \sum_g m_g$ and $G^{-1} \sum_g F_g$ — so sample-level point evaluation uses the target-exact plug-in weights $1/N$ (respondent) and $1/(G m_g)$ (equal-cluster) per respondent.

A.2. **The Weighted Polytomous Functional.** Items $j = 1, \dots, J$ carry declared, strictly increasing numeric score supports fixed before inference. For any distribution

F with margins F_j,

$$H(F) = \frac{\sum_{j<k} \operatorname{Cov}_F(Y_j, Y_k)}{\sum_{j<k} \operatorname{Cov}^{\max}(F_j, F_k)}, \qquad H_j(F) = \frac{\sum_{k\neq j} \operatorname{Cov}_F(Y_j, Y_k)}{\sum_{k\neq j} \operatorname{Cov}^{\max}(F_j, F_k)}, \tag{A.2}$$

with $\operatorname{Cov}^{\max}$ the fixed-margin maximum covariance (Appendix B). Aggregation is ratio-of-sums throughout — never the mean of pair coefficients — and the functional requires a positive scale denominator. Item selection, deletion, and re-keying are prohibited by the frozen estimand contract; on binary supports every quantity reduces exactly to the classical binary coefficient.

A.3. **Regularity and Knots.** For an item pair with weighted margins F_j, F_k, the comonotone transport path is determined by the interleaving of the two cumulative margins over the support points. A configuration is *regular* when all interior cumulative values are strictly separated (the path is locally constant under perturbation); it is an *exact knot* when interior cumulative margins coincide (the optimal transport table can switch under arbitrarily small perturbation, without changing the attained maximum at the point itself); *near-knot* configurations have separations below a frozen diagnostic threshold. Knot status is a property of the weighted margins — hence of the target weighting, as the SWMDK application illustrates — and is disclosed, by frozen contract, as a diagnostic quantity: no ordinary first-order inference is claimed at exact knots. The simulation's regularity lanes are constructed mechanically: distinct discrimination multipliers separate the weighted CDFs (regular lane); equating the first two, or all, item discriminations drives the margins toward or onto coincidence (near- and exact-knot lanes).

Appendix B. Transport, Weight-Law Identities, and Calibration Algebra

B.1. **The Denominator as a Derived Specialization.** For random variables with fixed margins, Rüschendorf (1982, p. 624, eq. 2) establishes that the common-quantile (comonotone) coupling $X_j^* = F_j^{-1}(U)$, $X_k^* = F_k^{-1}(U)$ with a shared uniform U maximizes $\mathbb{E}\,\psi(X_j + X_k)$ over all joint laws with those margins, for convex ψ.

Proposition B.1 (Specialization, not a new theorem)**.** *With two variables, $\psi(s) = s^2$, and finite second moments, the comonotone coupling maximizes* $\operatorname{Cov}(X_j, X_k)$ *among joint laws with margins F_j, F_k.*

Proof. $\mathbb{E}(X_j+X_k)^2 = \mathbb{E}X_j^2+\mathbb{E}X_k^2+2\mathbb{E}X_jX_k$. The marginal second moments are fixed by F_j and F_k, so maximizing the convex-sum expectation is equivalent to maximizing $\mathbb{E}X_jX_k$, hence (fixed means) the covariance. □

On finite ordered supports with arbitrary nonnegative masses, the comonotone coupling is computed exactly by the increasing two-pointer transport: initialize both margins' remaining masses at their lowest support points; repeatedly allocate the minimum of the two current remaining masses to the current support pair, accumulate the cross-moment, and advance whichever margin was exhausted. Every allocation exhausts at least one support point, so the loop performs at most $K + L - 1$ positive allocations for margin support sizes K and L, and constructs the discrete common-quantile coupling; pointers advance only on exactly exhausted mass (valid masses are shipped however small, never dropped by a tolerance), and the implementation asserts that the transported mass totals one. Cumulative ties (knots) permit non-unique tables with identical attained maxima. The implementation is cross-verified against an independent linear-programming solver on 25 frozen fixtures (maximum absolute discrepancy 1.4×10^{-12}; tolerance 2×10^{-7}), against 50 independent quantile-coupling fixtures, and against adversarial fixtures with valid masses at the 10^{-13} scale. By Molenaar (1991, Theorems 2 and 5), the covariance-ratio objects computed this way coincide with the weighted Guttman-error formulation of the multicategory coefficient; the archived formula–assumption bridge (seven verified rows) records the page-level correspondence. One scope note belongs here: Molenaar's presentation scores categories with consecutive integers, any equidistant scoring being equivalent; the arbitrary declared-score generality used in this article is a scope extension of the same covariance-ratio object, coinciding with his coefficient on consecutive scores, and the archived crosswalk records the difference explicitly.

B.2. **Exact Identities of the Two-Stage Law.** Let $(A_1, \ldots, A_G) \sim \mathrm{Dirichlet}_G(\mathbf{1})$ and, independently, $(B_{g1}, \ldots, B_{gm_g}) \sim \mathrm{Dirichlet}_{m_g}(\mathbf{1})$ within each cluster, with draw weights $W_{gi}^{(\mathrm{R})} = m_g A_g B_{gi} / \sum_h m_h A_h$ and $W_{gi}^{(\mathrm{C})} = A_g B_{gi}$.

Proposition B.2 (Weight-law identities)**.** *For every draw: (a) all weights are nonnegative and $\sum_{g,i} W_{gi} = 1$ exactly, under either target; (b) replacing every Dirichlet vector by its mean returns exactly the target's plug-in weighting — $W_{gi}^{(\mathrm{R})} \mapsto 1/N$ and $W_{gi}^{(\mathrm{C})} \mapsto 1/(Gm_g)$; (c) the law is invariant under relabeling of clusters and of respondents within clusters; (d) $m_g = 1$ clusters and unequal sizes are exact special cases (for $m_g = 1$, $B_{g1} = 1$ almost surely).*

Proof. (a) For the equal-cluster target, $\sum_{g,i} A_g B_{gi} = \sum_g A_g \sum_i B_{gi} = \sum_g A_g = 1$; the respondent target normalizes explicitly by $\sum_h m_h A_h$, and $\sum_{g,i} m_g A_g B_{gi} = \sum_g m_g A_g$. (b) $\mathbb{E} A_g = 1/G$ and $\mathbb{E} B_{gi} = 1/m_g$; substitution gives $m_g(1/G)(1/m_g)/\{\sum_h m_h/G\} =$

$1/N$ and $(1/G)(1/m_g)$, respectively. (c) Dirichlet($\mathbf{1}$) vectors are exchangeable and the two stages are independent, so any relabeling induces the same joint law. (d) is immediate from the Dirichlet definition. $\square$

These identities — plus reproducible-seed equality across batch sizes and worker counts — are each enforced by dedicated tests in the verification suite, so the algebra above is also a machine-checked property of the shipped implementation.

B.3. **Exact Properties of the Shrinkage Map.** Let $[L_n, U_n]$ be a defined raw equal-tail interval, m_n its midpoint, w_n its half-width, and $[\widetilde{L}_n, \widetilde{U}_n] = [m_n - k_G w_n, m_n + k_G w_n]$ with $k_G = 1 + \gamma/\sqrt{G}$ and $\gamma \in (-\sqrt{G}, 0\,]$.

Proposition B.3 (Exact finite-sample properties)**.** *(a) The midpoint is unchanged; (b) the length is multiplied by exactly $k_G \in (0, 1]$; (c) $[\widetilde{L}_n, \widetilde{U}_n] \subseteq [L_n, U_n]$, so for every θ, $\mathbf{1}\{\theta \in [\widetilde{L}_n, \widetilde{U}_n]\} \leq \mathbf{1}\{\theta \in [L_n, U_n]\}$: on any replication set the calibrated coverage is at most the raw coverage.*

Proof. $0 < k_G \leq 1$ gives $m_n - w_n \leq m_n - k_G w_n$ and $m_n + k_G w_n \leq m_n + w_n$; (b) is $2k_G w_n / 2w_n$. $\square$

This is the mirror image of a widening calibration: shrinkage can only reduce coverage, and it is justified precisely when the raw procedure demonstrably overcovers, which is the situation that the pilot, the held-out validation, and the fresh confirmation all establish for the two-stage law.

Lemma B.1 (Cluster-rate perturbation)**.** *For fixed γ, the endpoint changes are exactly $\mp\gamma w_n/\sqrt{G}$. If $w_n = O_p(G^{-1/2})$, they are $O_p(G^{-1})$ and $\sqrt{G}\,(\widetilde{L}_n - L_n) = o_p(1)$, likewise for the upper endpoint.*

Corollary B.1 (First-order limit preservation)**.** *If $(\sqrt{G}\{L_n - \theta\}, \sqrt{G}\{U_n - \theta\})$ converges jointly in distribution with no limit mass on the coverage boundary, the calibrated endpoints share the same limit and the two coverages converge to a common value.*

Proof. As in the widening case of Lee (2026b): Slutsky for the joint limit; continuity of the coverage functional at the limit law under the boundary condition. $\square$

Remark B.1 (What is not proved)**.** We do not assert a conditional limit theorem showing that the two-stage law's draw distribution matches the first-order sampling law of the clustered functional (the analog of Lo, 1987 for the independent-respondent

BB). Corollary B.1's premise — a $\sqrt{G}$-regular raw interval — is an assumption motivated by the cluster-rate design, supported by the numerical oracles, and stress-tested by the locked simulation chain. The overcoverage of the raw law at finite G, and the adequacy of a constant γ, are empirical findings under the frozen designs, not theorems.

Appendix C. Algorithm and Comparator Specifications

C.1. **The Locked Primary Procedure.** Method identifiers: `V4-P15-HBB-RESP-GN150-v1` (respondent target) and `V4-P15-HBB-CLUSTER-GN150-v1` (equal-cluster target), each the k_G transform of its raw base law. Given complete responses on the fixed item set with declared supports, cluster memberships, level .95, and B draws:

(1) Sort cluster labels by first appearance; draw $(A_g) \sim \mathrm{Dirichlet}_G(\mathbf{1})$ and, independently per cluster, $(B_{gi}) \sim \mathrm{Dirichlet}_{m_g}(\mathbf{1})$ from a seeded draw-major stream (Gamma(1,1) normalization) whose output is invariant to batch size and worker count.

(2) Form target weights $W_{gi}^{(\mathrm{R})} = m_g A_g B_{gi} / \sum_h m_h A_h$ (respondent) or $W_{gi}^{(\mathrm{C})} = A_g B_{gi}$ (equal-cluster; Appendix B); compute weighted margins and pairwise cross-moments; evaluate every pair denominator by the two-pointer transport; aggregate ratio-of-sums. Draws with nonpositive weighted scale denominator are typed invalid.

(3) If at least 99% of draws (and the frozen minimum count) are defined, the raw interval is the type-8 equal-tail $(.025, .975)$ quantile pair; otherwise the replication is typed undefined.

(4) Report $[m_n - k_G w_n, m_n + k_G w_n]$, $k_G = 1 - 1.5/\sqrt{G}$; no clipping; inherited typed status; no per-replication cascade. Undefined intervals count as misses in every performance denominator.

C.2. **Comparator Registry.** All comparators run on identical replications with identical failure accounting (frozen weight-law specification):

Table C.1. Frozen Comparator Specifications

Procedure (ID)	Exact specification
iid respondent BB (`V4-CMP-IID-RESP-BB-v1`)	Negative control for the working-independence respondent target: $(W_1, \ldots, W_N) \sim \mathrm{Dirichlet}_N(\mathbf{1})$ ignoring clusters; type-8 equal-tail quantiles. The frozen specification prohibits labeling this procedure cluster-valid.
One-stage cluster BB (`V4-CMP-ONE-STAGE-CLUSTER-BB-RESP-v1`)	Cluster weights only: $W_{gi} = A_g / \sum_h m_h A_h$ with within-cluster empirical distributions fixed; respondent target.
Two-stage frequentist bootstrap (`V4-CMP-TWO-STAGE-FREQ-BOOT-RESP-v1`)	Resample G clusters with replacement, then m_g respondents with replacement within each selected cluster; percentile-type interval from type-8 quantiles; respondent target.
Two-level delta Wald (`V4-CMP-MOKKEN-TWOLEVEL-DELTA-v1`)	$\widehat{H} \pm z_{.975}$ times the analytic two-level standard error in the lineage of `mokken`'s `MLcoefH` (Koopman, Zijlstra, & van der Ark, 2020); respondent-weighted working target.
Within-cluster HBB (`V4-DIAG-WITHIN-STAGE-HBB-RESP-v1`)	Diagnostic decomposition: within-cluster Dirichlet weights only, cluster weights fixed at their means; respondent target.

C.3. **Typed Failure Taxonomy.** Each replication carries one typed status per procedure, distinguishing at minimum: success; nonpositive weighted (or resampled) scale denominator; insufficient defined draws (the 99%/minimum-count rule); undefined analytic standard error (delta only); and nonfinite or reversed endpoints. Statuses are stored in the raw draw files, audited by the evaluators, never silently converted, and the exact vocabulary is part of the hash-locked protocol in the replication archive. Across the entire fresh confirmation and the SWMDK application, no procedure produced a single invalid replication or draw beyond the typed accounting (maximum invalid fraction 0).

Appendix D. Data-Generating Process, Truth Oracle, and Designs

D.1. **The Clustered Graded Mechanism.** Respondents answer $J = 5$ items with five ordered categories (scores 0–4) under a graded cumulative-logit model (Samejima, 1969) with conditional independence given

$$\theta_{gi} = \sqrt{\mathrm{ICC}}\, u_g + \sqrt{1 - \mathrm{ICC}}\, e_{gi}, \qquad u_g, e_{gi} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \tag{D.1}$$

so the latent intraclass correlation is the design's ICC parameter exactly. Items carry stage-frozen threshold profiles: development and held-out validation use six — standard, rare-high, and rare-low category variants plus shifted versions (e.g., standard $= (-1.20, -0.35, 0.40, 1.25)$; rare-high $= (-1.80, -0.80, 0.30, 2.10)$) — while the fresh confirmation uses three *confirm* variants disjoint from all of these (e.g., standard-confirm $= (-1.05, -0.28, 0.48, 1.33)$). Item location offsets and discrimination multipliers $\{.85, .925, 1, 1.075, 1.15\}$ around a cell-specific center complete the item side. Regularity lanes are constructed by discrimination ties: distinct multipliers (regular), first two equal with a small stage-frozen offset — 2×10^{-4} in training, 3.5×10^{-4} in validation, 5×10^{-4} in the confirmation — (near knot), all equal with zero offsets (exact knot).

Cluster sizes are bounded in $[5, 29]$ and generated by one of four mechanisms: balanced ($m_g = 20$); SWMDK-like (resampled with replacement from the observed SWMDK sizes); and informative-positive/negative ($m_g = 5 + \text{round}\{24\,\text{plogis}(\pm s\, u_g)\}$ with cell-specific slope s), which couple size to the cluster effect and thereby separate F_{R} from F_{C}. Replications resample cluster effects, sizes, and respondents — repeated sampling is unconditional over all three.

D.2. **Truth Oracle.** Population truth for both targets is computed by two-dimensional Gauss–Hermite quadrature over (u, e) — 41 nodes per dimension in development and held-out validation, 51 in the fresh confirmation, each frozen in its stage's protocol: category probabilities integrate to margins and pairwise joints; the respondent-weighted target weights each cluster-effect node by its size-mechanism's expected size, the equal-cluster target equally; denominators evaluate through the same exact transport as the estimator. No truth is ever obtained by simulation, and the finite-transport implementation is oracle-verified (Appendix B).

D.3. **The Three Designs.** *Training* (24 cells; 120 replications each; 2,880 scheduled; $B = 299$): crossing G, ICC, size mechanisms, discrimination centers, and lanes; used only to screen and select γ. *Held-out validation* (18 cells; 180 each; 3,240 scheduled; $B = 299$): conditions and seeds disjoint from training; used once, for the locked candidate and its prespecified fallback. *Fresh confirmation* (24 cells; 400 each; 9,600 scheduled per target; $B = 499$): newly parameterized after the final lock — $G \in \{22, 35, 55, 80\}$, ICC $\in \{.06, .12, .19, .26, .34, .38\}$, all four size mechanisms (informative slopes .75–1.30), fresh *-confirm* threshold profiles, 16 regular + 4 near-knot + 4 exact-knot cells (Table F.1 lists them with results). All heavy computation ran on 12 parallel cloud workers with per-cell checkpoints; raw outputs were downloaded in full and hash-manifested before any evaluation.

Appendix E. Chronology, Decisions, and Gates

E.1. **Pilot (Feasibility).** The feasibility pilot ran 1,080 replications across regular, near-knot, and exact-knot lanes with the full procedure battery. Raw two-stage HBB coverage was .994 overall (the overcoverage finding), spanning .992–.997 by lane on the primary respondent target and .989–1.000 on the equal-cluster sensitivity lanes; the working-independence BB covered .918 overall and the one-stage law .913, while the two-level delta (.981) and two-stage frequentist bootstrap (.985) sat high of nominal at greater length. The pilot fixed the diagnosis and the feasibility gate (G14) only; no calibration constant existed yet.

E.2. **Training and Selection.** Table E.1 reproduces the frozen training decisions. Candidates $\gamma \in \{0, -.5, -.75, -1\}$ failed eligibility (residual overcoverage against the $[.94, .98]/[.94, .985]$ bands); -1.25 and -1.5 were eligible; the prespecified minimum-length rule selected -1.5 (6.3% shorter on the primary target). The training method lock was serialized with zero validation outputs in existence (`validation_output_count_at_lock` = 0 in the machine-readable receipt).

TABLE E.1. Training-Stage Candidate Decisions Across the Frozen γ Grid

γ	Coverage		Mean length		Min G	Min mech.	Eligible
	Resp.	Eq.-cl.	Resp.	Eq.-cl.	group	group	
0.00	.9952	.9970	.1338	.1393	.993	.992	no
−0.50	.9905	.9911	.1210	.1260	.988	.983	no
−0.75	.9857	.9839	.1146	.1193	.980	.978	no
−1.00	.9815	.9768	.1083	.1127	.968	.969	no
−1.25	.9714	.9708	.1019	.1060	.960	.964	yes
−1.50	.9607	.9589	.0955	.0994	.938	.953	yes

E.3. **Held-Out Validation and the Failed Fallback.** Table E.2 shows the single evaluation of the locked candidate on the 18 disjoint cells: coverage .9718/.9722 inside the acceptance bands at lengths .088/.091, zero invalid replications. The prespecified fallback — revert to the raw interval if the candidate failed — was evaluated under its own frozen bands and failed them (.9949 and .9968, above the .99/.995 caps): the two negative rows in the archived gate table are the fallback's, by design. The final method lock then froze the procedure and the confirmation gates.

Table E.2. Held-Out Validation of the Locked Candidate (18 Disjoint Cells)

Method role	Target	Coverage (MCSE)	Mean length	Invalid
Raw ($\gamma = 0$)	equal_cluster	.9968 (.0012)	.1198	.000
Raw ($\gamma = 0$)	respondent	.9949 (.0015)	.1157	.000
Selected ($\gamma = -1.5$)	equal_cluster	.9722 (.0035)	.0912	.000
Selected ($\gamma = -1.5$)	respondent	.9718 (.0036)	.0881	.000

E.4. **Confirmation, Application, and Review Gates.** The fresh confirmation (Section 5) passed 7 of 7 blocking checks and issued `CONFIRM_CLUSTER_POLYTOMOUS_METHOD`; raw outputs were downloaded in full before evaluation, and the decision receipt pins the protocol, method-lock, manifest, and gate hashes. The SWMDK application required separate authority and passed 14 of 14 blocking criteria (G17-B01 to G17-B14), including provenance match to the frozen dataset digest, completeness of all 12 computation tasks and their hash chains, an invalid-fraction cap, point-estimate concordance below 10^{-10} against an independent implementation, split-half endpoint stability, mandatory reporting of both targets and every comparator, complete cluster/ICC/knot disclosures, and the required no-threshold-decision labeling. A subsequent source-review gate closed page-level crosswalks to the three original sources — Molenaar (1991) (8 questions), Rüschendorf (1982) (6), and Andreadis (2017) (7, file-and-line against the archived release) — verified the seven-row formula–assumption bridge, froze the claim-eligibility register of Appendix H, and passed 13 of 13 blocking criteria; the production gate that ported the method into the public package passed 19 of 19 with the binary path of Lee (2026b) certified bitwise unchanged. No scientific parameter, seed, gate, or acceptance rule changed after its stage's lock anywhere in this chronology.

Appendix F. Complete Confirmation Results

Table F.1 lists the selected method's unconditional coverage and mean length in every confirmation cell, both targets, with the design coordinates and lane; Figure F.1 displays the same information against the frozen bands and the .88 regular-cell floor. Because the locked phase-16 cell-summary artifact covers the 16 regular decision cells only, the all-lane table here is derived directly from the locked raw replication rows, and its regular rows are cross-checked against that artifact during the build. Across all 24 cells and both targets, coverage spans .925–.995; the all-lane minimum (.925) occurs in a near-knot diagnostic cell, and the regular-lane minimum is .930. Coverage rises from near-nominal at $G = 22$ toward mild conservatism at $G = 80$ (the expected consequence of a single cluster-rate constant), and no cell of any lane falls below the frozen floor. Table F.2 gives the full comparator-by-lane table (respondent target): the naive iid BB degrades from .915 (regular) to .880 (exact knot), the within-stage and one-stage decompositions undercover everywhere, and the conservative pair (two-level delta, two-stage bootstrap) remains above the band ceiling in every lane at the design's greatest lengths. Every procedure in every cell completed all scheduled replications with typed status success (maximum invalid fraction 0), so no coverage number in this supplement conditions on definedness.

TABLE F.1. Calibrated HBB: All Confirmation Cells by Target

Cell	Target	G	ICC	Lane	Coverage	MCSE	Length
P16-C001	equal_cluster	22	.06	regular	.9700	.0085	.0883
P16-C001	respondent	22	.06	regular	.9750	.0078	.0889
P16-C002	equal_cluster	22	.19	regular	.9475	.0112	.1117
P16-C002	respondent	22	.19	regular	.9525	.0106	.1052
P16-C003	equal_cluster	22	.34	near_knot	.9275	.0130	.1272
P16-C003	respondent	22	.34	near_knot	.9250	.0132	.1218
P16-C004	equal_cluster	22	.12	regular	.9525	.0106	.1206
P16-C004	respondent	22	.12	regular	.9475	.0112	.1162
P16-C005	equal_cluster	22	.26	exact_knot	.9475	.0112	.1089
P16-C005	respondent	22	.26	exact_knot	.9475	.0112	.1096
P16-C006	equal_cluster	22	.38	regular	.9325	.0125	.1079
P16-C006	respondent	22	.38	regular	.9300	.0128	.1035
P16-C007	equal_cluster	35	.06	regular	.9750	.0078	.1028
P16-C007	respondent	35	.06	regular	.9775	.0074	.0999
P16-C008	equal_cluster	35	.19	regular	.9775	.0074	.0967
P16-C008	respondent	35	.19	regular	.9800	.0070	.0968
P16-C009	equal_cluster	35	.34	regular	.9850	.0061	.0834
P16-C009	respondent	35	.34	regular	.9675	.0089	.0787
P16-C010	equal_cluster	35	.12	near_knot	.9825	.0066	.0884
P16-C010	respondent	35	.12	near_knot	.9850	.0061	.0884
P16-C011	equal_cluster	35	.26	exact_knot	.9350	.0123	.0973

Cell	Target	G	ICC	Lane	Coverage	MCSE	Length
P16-C011	respondent	35	.26	exact_knot	.9325	.0125	.0923
P16-C012	equal_cluster	35	.38	regular	.9675	.0089	.1172
P16-C012	respondent	35	.38	regular	.9650	.0092	.1103
P16-C013	equal_cluster	55	.06	regular	.9900	.0050	.0857
P16-C013	respondent	55	.06	regular	.9950	.0035	.0808
P16-C014	equal_cluster	55	.19	regular	.9850	.0061	.0753
P16-C014	respondent	55	.19	regular	.9850	.0061	.0725
P16-C015	equal_cluster	55	.34	regular	.9725	.0082	.0871
P16-C015	respondent	55	.34	regular	.9725	.0082	.0869
P16-C016	equal_cluster	55	.12	near_knot	.9775	.0074	.0968
P16-C016	respondent	55	.12	near_knot	.9900	.0050	.0904
P16-C017	equal_cluster	55	.26	exact_knot	.9675	.0089	.0897
P16-C017	respondent	55	.26	exact_knot	.9650	.0092	.0867
P16-C018	equal_cluster	55	.38	regular	.9725	.0082	.0919
P16-C018	respondent	55	.38	regular	.9725	.0082	.0850
P16-C019	equal_cluster	80	.06	regular	.9875	.0056	.0812
P16-C019	respondent	80	.06	regular	.9900	.0050	.0767
P16-C020	equal_cluster	80	.19	regular	.9825	.0066	.0655
P16-C020	respondent	80	.19	regular	.9875	.0056	.0654
P16-C021	equal_cluster	80	.34	near_knot	.9650	.0092	.0808
P16-C021	respondent	80	.34	near_knot	.9700	.0085	.0771
P16-C022	equal_cluster	80	.12	regular	.9900	.0050	.0778
P16-C022	respondent	80	.12	regular	.9875	.0056	.0759
P16-C023	equal_cluster	80	.26	exact_knot	.9725	.0082	.0726
P16-C023	respondent	80	.26	exact_knot	.9750	.0078	.0723
P16-C024	equal_cluster	80	.38	regular	.9925	.0043	.0635
P16-C024	respondent	80	.38	regular	.9925	.0043	.0602

Figure F.1. Selected calibrated HBB: coverage in every confirmation cell (both targets), ordered by number of clusters and latent ICC. Shading: frozen respondent decision band; dashed: nominal .95; dotted: frozen regular-cell floor .88. Fill distinguishes regularity lanes; error bars are ±1 MCSE at 400 replications per cell.

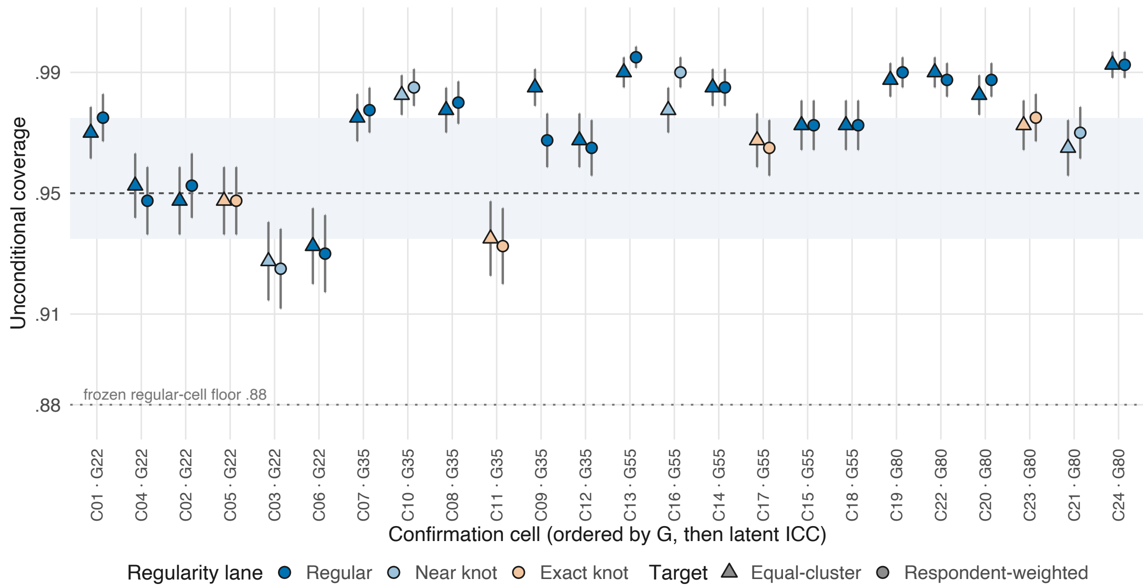


Table F.2. Comparator Procedures by Regularity Lane (Respondent Target)

Procedure	Lane	Coverage (MCSE)	Mean length	Invalid
Two-level delta (analytic)	exact_knot	.9725 (.0041)	.1002	.000
Two-level delta (analytic)	near_knot	.9669 (.0045)	.1020	.000
Two-level delta (analytic)	regular	.9809 (.0017)	.0939	.000
Two-stage cluster NPB	exact_knot	.9800 (.0035)	.1059	.000
Two-stage cluster NPB	near_knot	.9844 (.0031)	.1106	.000
Two-stage cluster NPB	regular	.9897 (.0013)	.1012	.000
One-stage cluster BB	exact_knot	.9313 (.0063)	.0790	.000
One-stage cluster BB	near_knot	.9213 (.0067)	.0812	.000
One-stage cluster BB	regular	.9261 (.0033)	.0734	.000
iid respondent BB (naive)	exact_knot	.8800 (.0081)	.0690	.000
iid respondent BB (naive)	near_knot	.8994 (.0075)	.0738	.000
iid respondent BB (naive)	regular	.9148 (.0035)	.0691	.000
Within-cluster HBB (diagnostic)	exact_knot	.8612 (.0086)	.0650	.000
Within-cluster HBB (diagnostic)	near_knot	.8706 (.0084)	.0692	.000
Within-cluster HBB (diagnostic)	regular	.8956 (.0038)	.0652	.000

Appendix G. SWMDK Application: Details

G.1. **Provenance and Fixed Scales.** `SWMDK` ships with the `mokken` package as a documented subset of the Dutch COOL5-18 cohort study (Zijsling et al., 2017a, 2017b): 639 pupils in 30 classrooms, 13 five-category items from the school-well-being scales of Peetsma et al. (2001), subset construction per Koopman et al. (2022). The frozen application contract fixed, in advance: the two analysis scales — well-being with teachers (Items 1–6) and well-being with classmates (Items 8, 9, 10, 11, 13) — the classroom identifier as cluster, both targets, the seeds, all comparators, and the gates; item reselection and re-keying were prohibited; the dataset digest was locked (SHA-256 `e1257f...`) and re-verified at closure. Observed cluster sizes span 5–29 (quartiles 18.25/23/27); intraclass correlations are .169 (teachers) and .183 (classmates), both with $p < 10^{-15}$ in the one-way decomposition. Every fixed item retains all five observed categories; Table G.1 gives the full category profile (smallest single-category mass .0031, count 2).

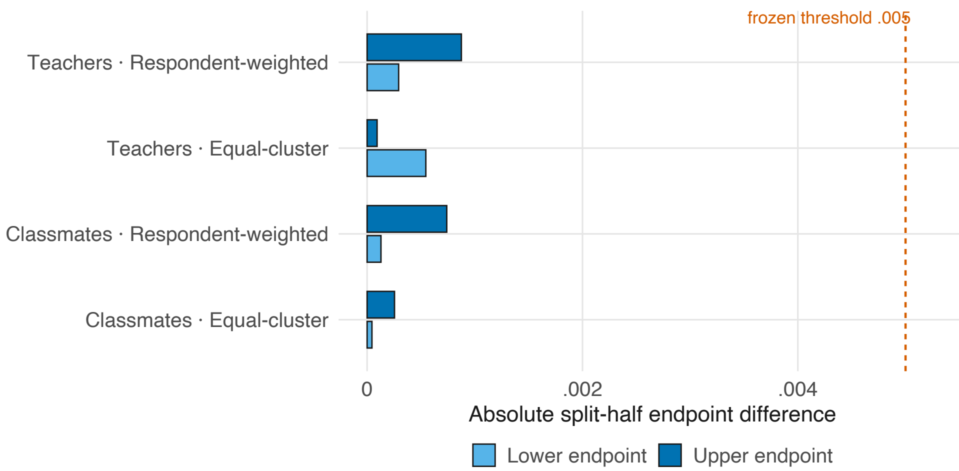
Table G.1. SWMDK Fixed-Item Category Profile

Scale	Item	Score	Count	Mass
classmate	Item10	1	16	.0250
classmate	Item10	2	31	.0485
classmate	Item10	3	143	.2238
classmate	Item10	4	265	.4147
classmate	Item10	5	184	.2879
classmate	Item11	1	4	.0063
classmate	Item11	2	10	.0156
classmate	Item11	3	113	.1768
classmate	Item11	4	358	.5603
classmate	Item11	5	154	.2410
classmate	Item13	1	2	.0031
classmate	Item13	2	14	.0219
classmate	Item13	3	123	.1925
classmate	Item13	4	343	.5368
classmate	Item13	5	157	.2457
classmate	Item8	1	2	.0031
classmate	Item8	2	20	.0313
classmate	Item8	3	92	.1440
classmate	Item8	4	347	.5430

Scale	Item	Score	Count	Mass
classmate	Item8	5	178	.2786
classmate	Item9	1	21	.0329
classmate	Item9	2	49	.0767
classmate	Item9	3	137	.2144
classmate	Item9	4	207	.3239
classmate	Item9	5	225	.3521
teacher	Item1	1	55	.0861
teacher	Item1	2	137	.2144
teacher	Item1	3	307	.4804
teacher	Item1	4	133	.2081
teacher	Item1	5	7	.0110
teacher	Item2	1	30	.0469
teacher	Item2	2	99	.1549
teacher	Item2	3	261	.4085
teacher	Item2	4	218	.3412
teacher	Item2	5	31	.0485
teacher	Item3	1	49	.0767
teacher	Item3	2	116	.1815
teacher	Item3	3	269	.4210
teacher	Item3	4	173	.2707
teacher	Item3	5	32	.0501
teacher	Item4	1	14	.0219
teacher	Item4	2	27	.0423
teacher	Item4	3	248	.3881
teacher	Item4	4	312	.4883
teacher	Item4	5	38	.0595
teacher	Item5	1	18	.0282
teacher	Item5	2	65	.1017
teacher	Item5	3	329	.5149
teacher	Item5	4	199	.3114
teacher	Item5	5	28	.0438
teacher	Item6	1	14	.0219
teacher	Item6	2	66	.1033
teacher	Item6	3	272	.4257
teacher	Item6	4	255	.3991
teacher	Item6	5	32	.0501

G.2. **Computation and Stability.** Twelve computation tasks (2 scales $\times$ 2 targets for the selected method's raw bases, plus the comparator batteries) each generated 99,999 seeded weight draws — 1,199,988 draws, zero invalid; the four selected intervals are deterministic k_G transforms of their raw draw streams, with the observed selected-to-raw length ratio matching $k_G = .7261387$ to within 4×10^{-15} across all four. The frozen split-half diagnostic recomputes each selected interval on the two half-streams: the largest absolute endpoint difference was .000875 against the .005 threshold (Figure G.1), and point-estimate concordance with an independent implementation was below 10^{-10} for every scale–target combination.

FIGURE G.1. Split-half endpoint stability of the four selected SWMDK intervals against the frozen .005 threshold.



G.3. **Knot Diagnostics by Target.** Under the respondent-weighted target both scales sit exactly at cumulative knots (two exact and two near knots each; minimum internal CDF gap 0); under the equal-cluster target the same responses produce none (minimum gaps .00121 and .000617). The disclosure is diagnostic by contract: the coefficients are exactly computed either way, the phenomenon illustrates that target weighting changes the tie structure of weighted margins, and the confirmation's exact-knot lane (.955–.956 selected-method coverage) bounds the inferential cost of such configurations within the tested design. No regular-inference claim is attached to the knot cases.

Appendix H. Software, Reproducibility, and the Claim Register

H.1. **The bayesmokken Clustered-Ordinal Path.** The public entry point is `bb_mokken_h()`:

```r
library(bayesmokken)
fit <- bb_mokken_h(data, item_type = "ordinal", supports = supports,
                   cluster = class_id,
                   cluster_target = "respondent",    # or "equal_cluster"
                   weight_law = "hierarchical_cluster",
                   draws = 2000, seed = 1, interval = "calibrated")
fit$interval   # method id, target, law, point, raw + calibrated
               # endpoints, k_G, draw counts, denominator and knot
               # diagnostics, typed status
```

Design decisions keep this call identical to the validated method. The population target and the item score supports are required arguments with no defaults, because defaulting either would let the software select an estimand or substitute the sample's categories for the instrument's; the confidence level is fixed at .95, the only level with calibration evidence; γ is not a user argument; and undefined states are returned as typed statuses, never silently repaired. The returned object records the method identifier, target, weight law, point estimate, raw and calibrated endpoints, k_G, draw counts, denominator and knot diagnostics, and a typed status, and target and law survive serialization, so an archived object is self-describing. At the production gate the package's suite ran 791 expectations across 24 files with zero failures, warnings, or skips — transport-versus-LP and quantile-fixture oracles, the weight-law identities of Appendix B, target-mass separation, RNG determinism across batch sizes and worker counts, serialization round trips, and unsupported-combination errors — and `R CMD check` completed cleanly.

H.2. **Scope of the Installed Package Versus This Article.** The installed package is broader than the procedure evaluated here, and the difference should be stated so that availability is not mistaken for validation. Alongside the clustered-ordinal path above, the package implements the independent-respondent binary method of Lee (2026b), with its own estimand, weight law, tuning constant of opposite sign, and validation chain; nothing in the present theory, simulation, or application bears on it. The package enforces the boundary rather than guessing: dichotomous items with a cluster argument are refused on every route, ordinal items under the independent-respondent law are refused, cluster-free ordinal random-weight intervals are refused, and the calibrated clustered interval is confined to its evidence envelope — it refuses fewer than 22 complete-scale clusters, and fits above 80 clusters carry an explicit extrapolation label (`EXTRAPOLATED_CLUSTER_COUNT`). At the production gate a frozen regression suite certified the binary path bitwise unchanged. For reproducing this article, the clustered-ordinal call above and its method identifiers (`V4-P15-HBB-RESP-GN150-v1`, `V4-P15-HBB-CLUSTER-GN150-v1`; Appendix C) are the complete contract.

H.3. **Release History Relevant to This Article.** The version evaluated at the production gate was 0.0.0.9000, and the archived tarball remains the version of record for the locked evidence; no claim attaches to later releases. Two corrections made after the gate are disclosed for completeness, neither affecting any number in this article. An external review found (a) that the deterministic per-pair transport kernel in the archived tarball advanced its pointers on a 10^{-12} mass tolerance, so a valid positive mass at or below that scale could be dropped, and (b) that a fractional seed passed to the draw-stream constructor was silently truncated rather than rejected. Version 0.0.0.9001 fixed both — pointers now advance only on exactly exhausted mass, transported-mass conservation is asserted, and adversarial tiny-mass fixtures were added. The batched survival-sum kernel used by every Bayesian-bootstrap draw in this article is byte-identical in both versions. The public release at the time of writing is 0.9.0, which adds a documented user-facing surface (data screening, result extraction, diagnostics, reporting, and figures) while carrying the estimation engine over unchanged: the release's parity tests compare the frozen 0.0.0.9001 numerical receipts field by field at zero tolerance, including all draw arrays; the calibration constants and the 95% level remain unreachable from the public interface; the cluster-count envelope guard described above was added in this release; and the call forms printed in this article run unchanged.

H.4. **Repositories.**

- Package: https://github.com/joonho112/bayesmokken
- Replication archive: https://github.com/joonho112/bayesmokken-clustered-polytomous-replication

The replication archive rebuilds every figure and table of this article and its supplement from the locked artifacts under hash manifests; the archived raw outputs, including the application's complete draw streams, make re-running the original computation unnecessary, and every random stream is seeded and draw-major, so results are independent of batch size and worker count.

H.5. **The Frozen Claim Register.** Table H.1 transcribes the claim-eligibility register frozen at the source-review gate. It is reproduced in full — allowed claims with their boundaries, diagnostic-only classifications, and forbidden claims — because it is the document against which the wording of the main text was audited, and it is reproduced as frozen, in the internal vocabulary of the research program: *v4* designates the program whose locked artifacts this article reports, *Phase 16* its confirmation stage (Section 5), and *the archived predecessor* the weighted implementation of Andreadis (2017).

TABLE H.1. The Frozen Claim-Eligibility Register Governing This Article

ID	Claim	Boundary	Status
CL-01	The standard weighted multicategory Loevinger H is a covariance-to-fixed-margin-maximum-covariance ratio, aggregated as a quotient of sums.	Use for finite ordered scores and positive denominators; do not imply inferential calibration.	ELIGIBLE
CL-02	The common-quantile coupling supplies the maximum pair covariance for v4 finite ordinal margins.	Describe as an established result and derived specialization, not a new transport theorem.	ELIGIBLE
CL-03	Weighted Mokken software predates v4.	Give explicit predecessor credit.	REQUIRED CREDIT
CL-04	v4 implements arbitrary empirical-mass transport and cluster-aware hierarchical random-weight inference for fixed polytomous scales.	No first-ever or universal-superiority wording.	ELIGIBLE
CL-05	Within the frozen Phase 16 design, the locked hierarchical method met all protocol gates.	Restrict to the tested finite design and method lock.	ELIGIBLE RESTRICTED
CL-06	In the frozen SWMDK application, cluster-aware uncertainty was empirically relevant for the prespecified scales and targets.	Descriptive application evidence; no population threshold decision.	ELIGIBLE RESTRICTED
CL-07	Respondent-weighted and equal-cluster targets are distinct estimands under informative cluster size.	Always label target and hierarchical weight law.	ELIGIBLE RESTRICTED
CL-08	Exact or near cumulative knots identify possible nonsmooth transport-path behavior.	Do not present as regular first-order inference or a failure of the H value itself.	DIAGNOSTIC ONLY
CL-09	v4 is the first weighted Mokken implementation.	Directly contradicted by the archived predecessor.	FORBIDDEN

ID	Claim	Boundary	Status
CL-10	v4 proves a new comonotone transport theorem.	The project implements and specializes an established result.	FORBIDDEN
CL-11	The selected hierarchical method is universally superior to all comparators.	Evidence is design- and comparator-specific.	FORBIDDEN
CL-12	The empirical H values establish a formal scale-quality threshold decision.	Threshold decisions were excluded.	FORBIDDEN
CL-13	Exact-knot cases have ordinary regular inference without qualification.	Exact knots remain diagnostic-only.	FORBIDDEN
CL-14	Rüschendorf (1982) establishes cluster sampling or Bayesian calibration.	The source concerns fixed-margin extremal dependence, not v4 inference.	FORBIDDEN